



自然语言处理

8. Transformer与预训练模型

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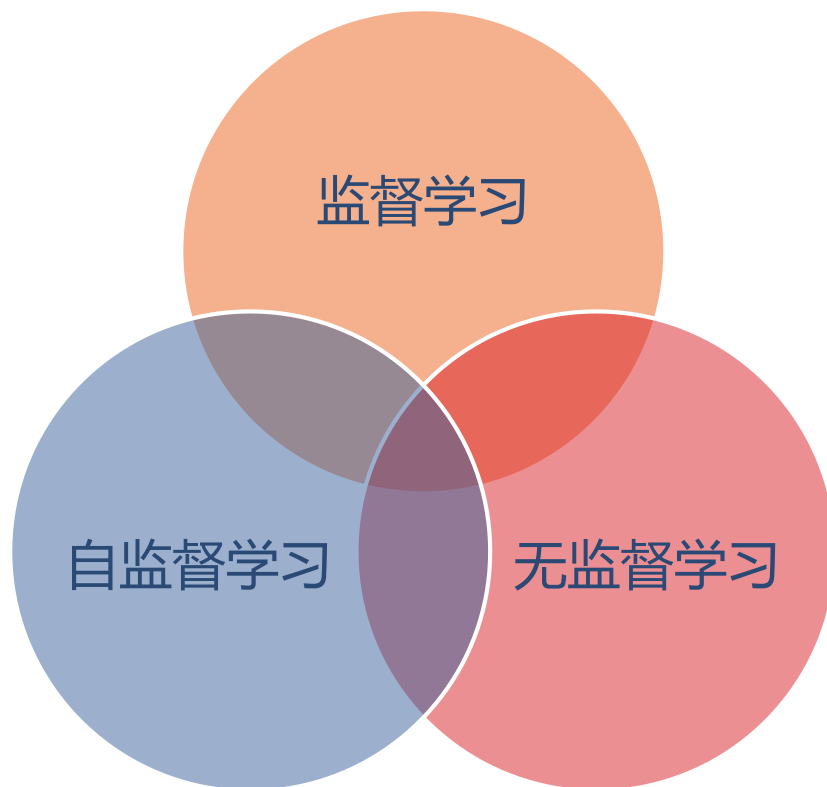
本章内容

- 8.1 引言
- 8.2 自注意力机制
- 8.3 Transformer网络架构
- 8.4 预训练模型及其应用
 - 8.4.1 基于Transformer编码器的预训练语言模型
 - 8.4.2 基于Transformer的预训练Seq2Seq模型
 - 8.4.3 基于Transformer解码器的预训练语言模型
 - 8.4.4 基于Transformer的文本-视觉预训练模型
- 8.5 本章小结

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预训练任务



预训练任务

- 自监督学习
 - 一种监督学习和无监督学习的结合
 - 监督式的学习方式
 - 训练数据由无标注数据自动构建
 - 典型任务
 - Language Modeling (LM)
 - CBoW, Skip-Gram, GloVe
 - Masked Language Modeling (MLM)
 - Next Sentence Prediction (NSP)
 - Masked Span Prediction
 -

预训练任务

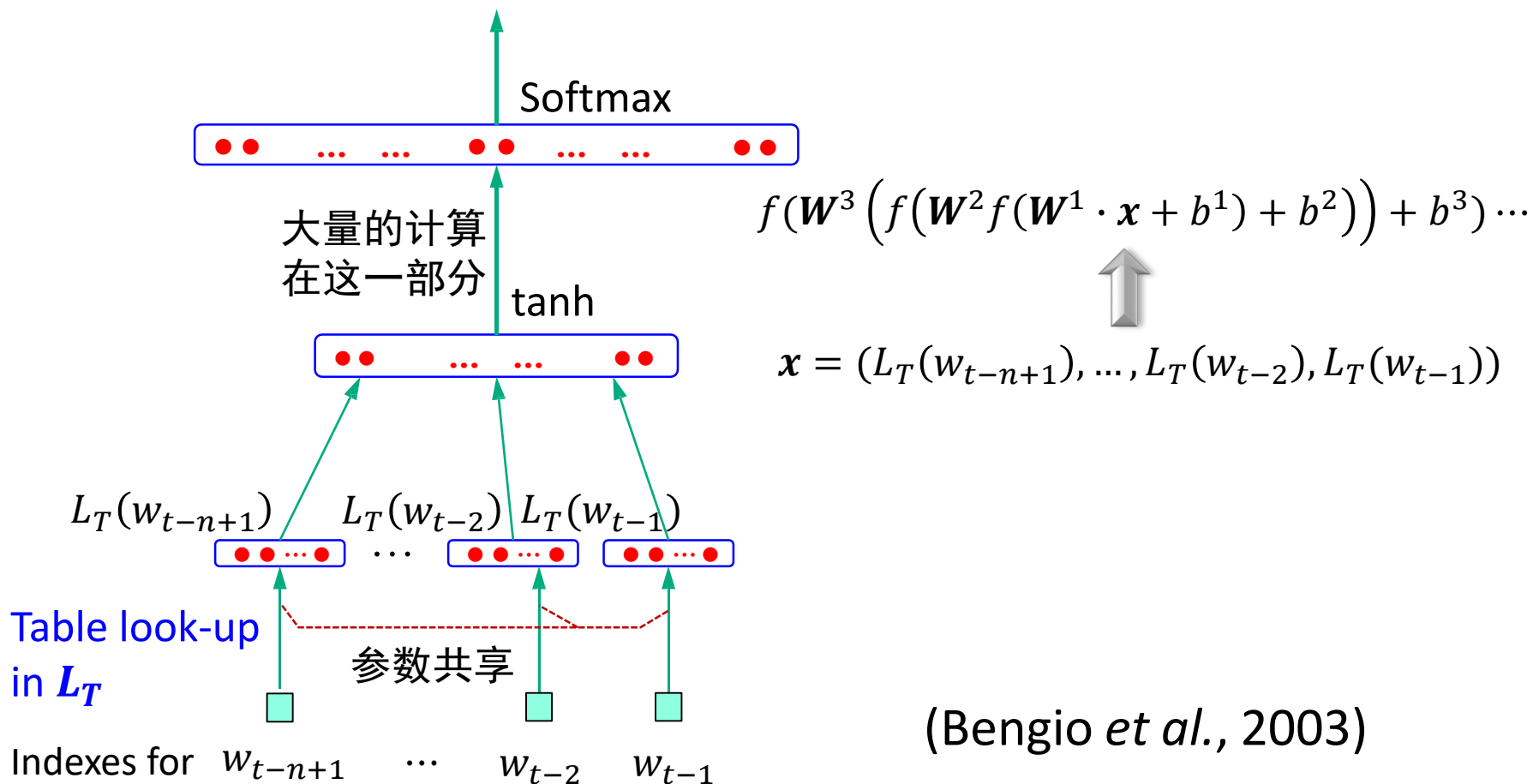
- 回顾前向神经网络神经网络语言模型
 - 给定第t个词的前n-1个词，最大化第t个词的似然估计

$$p(w_t | w_{t-n+1}^{t-1}) = f(w_t | w_{t-n+1}^{t-1}) = \frac{c(w_{t-n+1}^t)}{\sum_{w_t} c(w_{t-n+1}^t)}$$

预训练任务

- 回顾前向神经网络神经网络语言模型

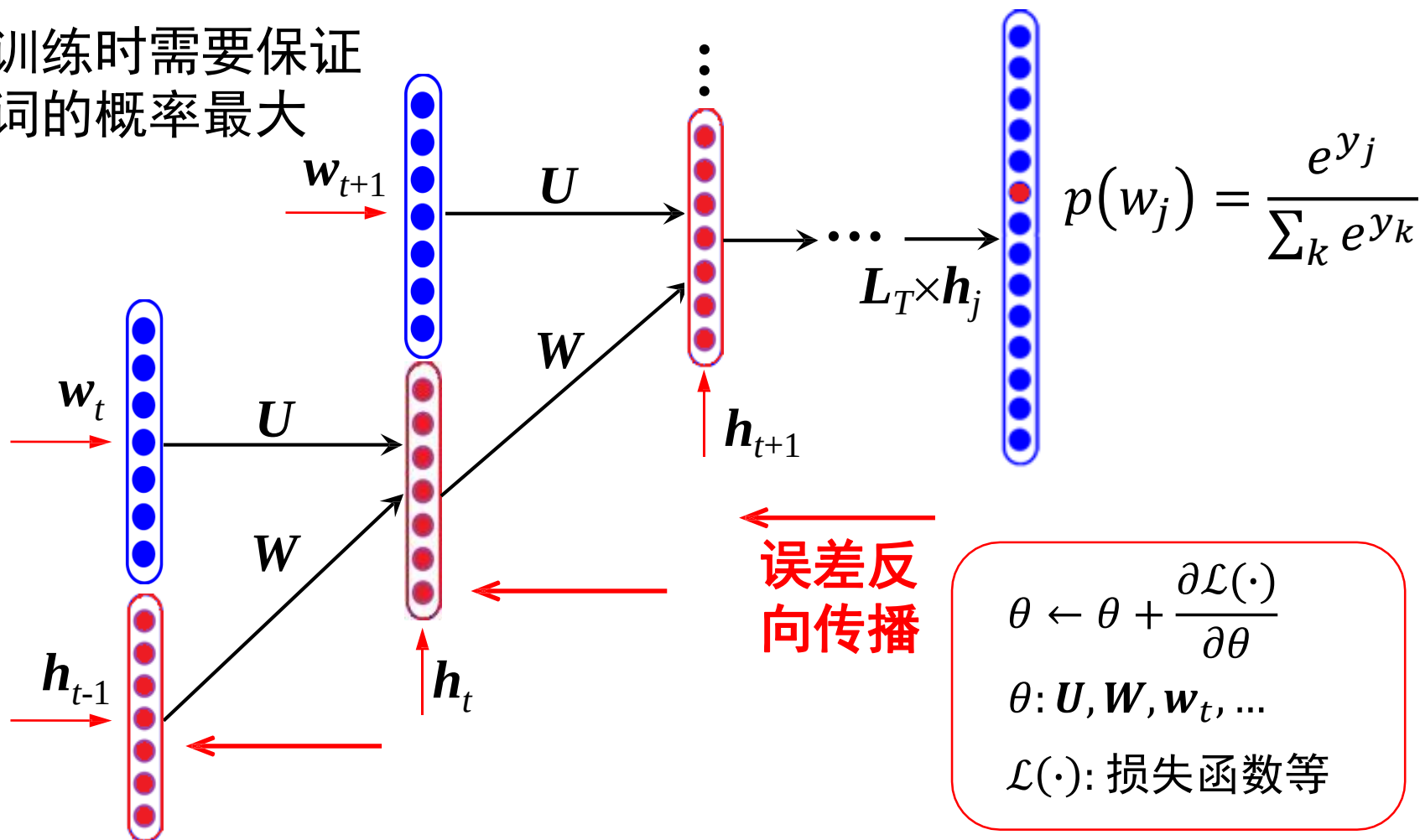
Output: $w_t = p(w_t | context)$



预训练任务

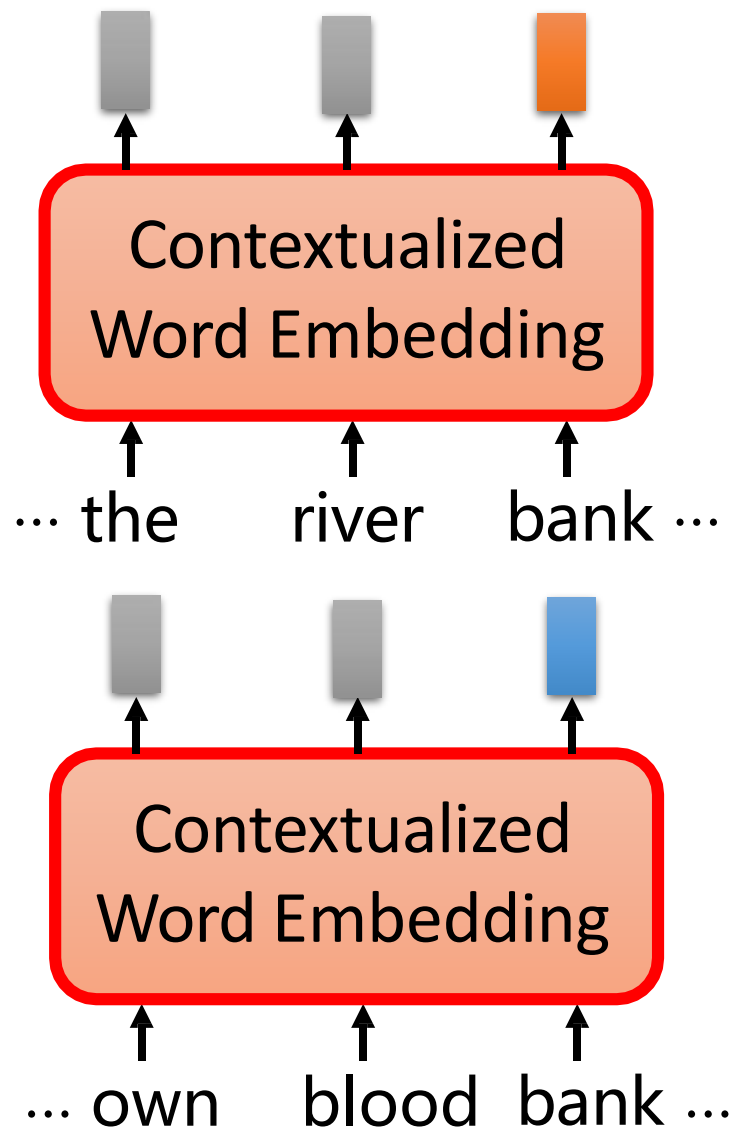
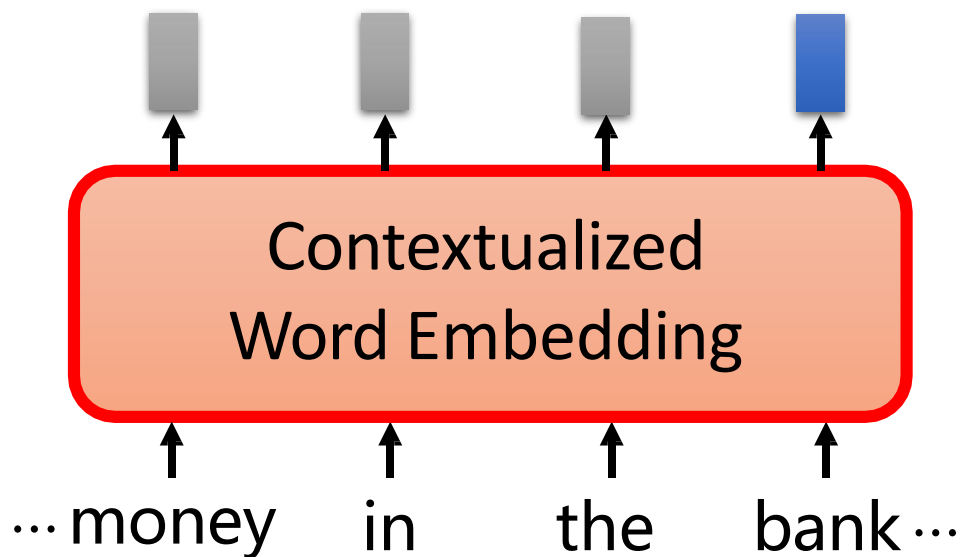
- 回顾循环神经网络神经网络语言模型

模型训练时需要保证
当前词的概率最大



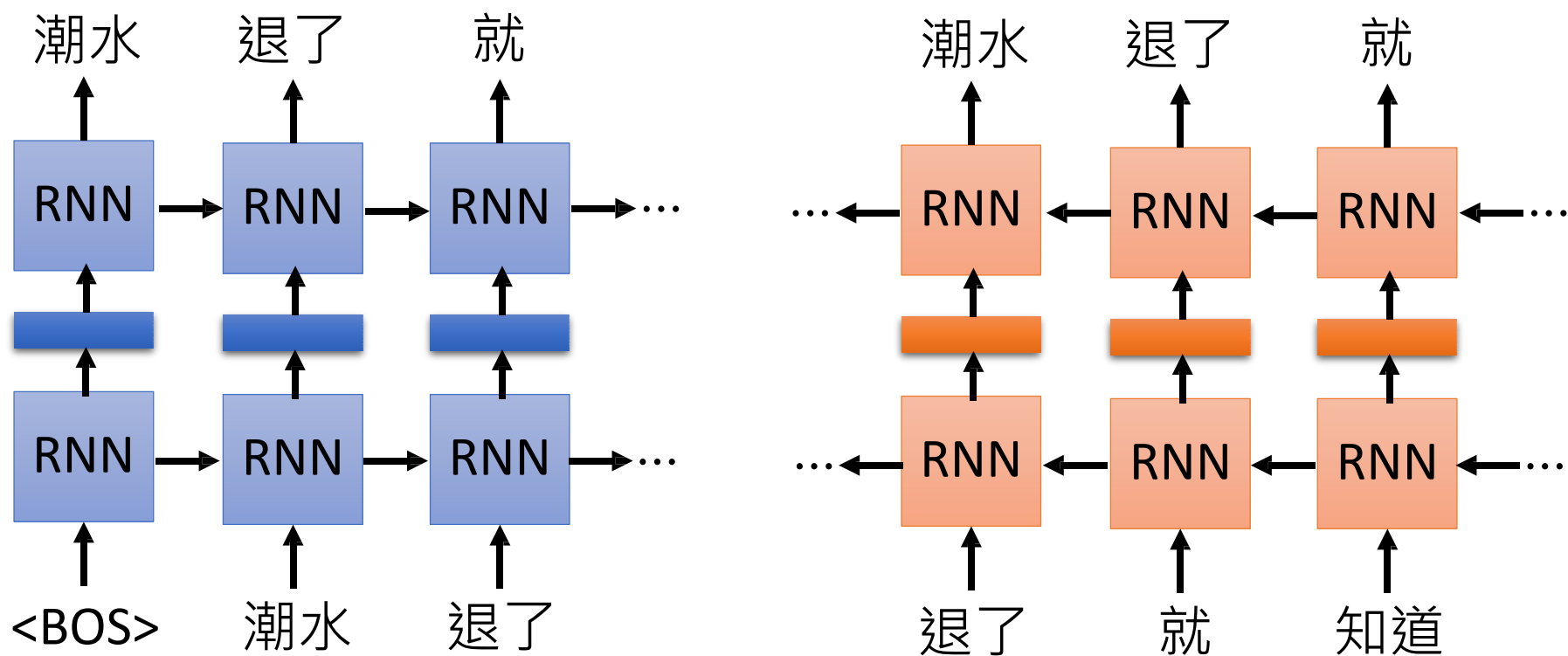
预训练任务

- 之前神经网络神经语言模型的缺陷
 - 无法解决一词多义问题



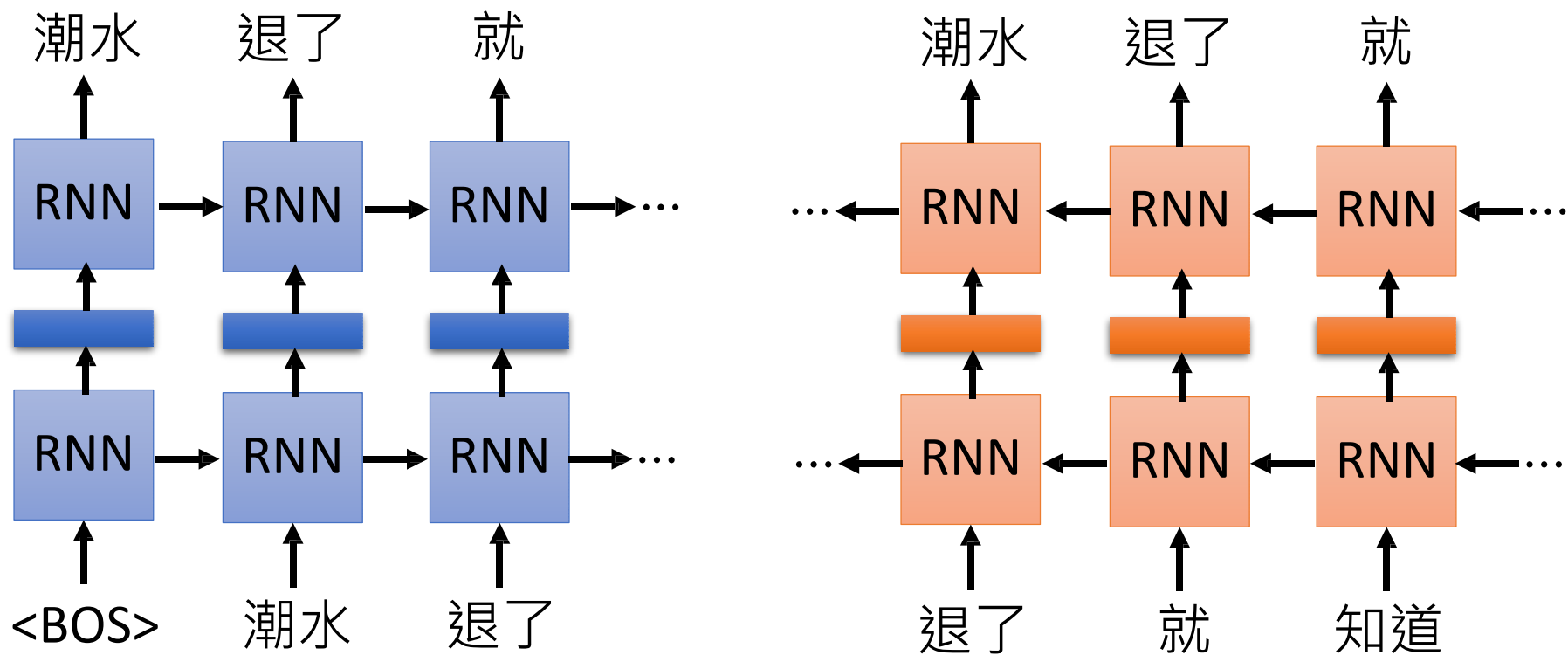
预训练任务

- Embeddings from Language Model (ELMO) (NAACL 2018 Best Paper)
 - RNN-based language models (trained from lots of sentences)
 - Bidirectional Modeling



预训练任务

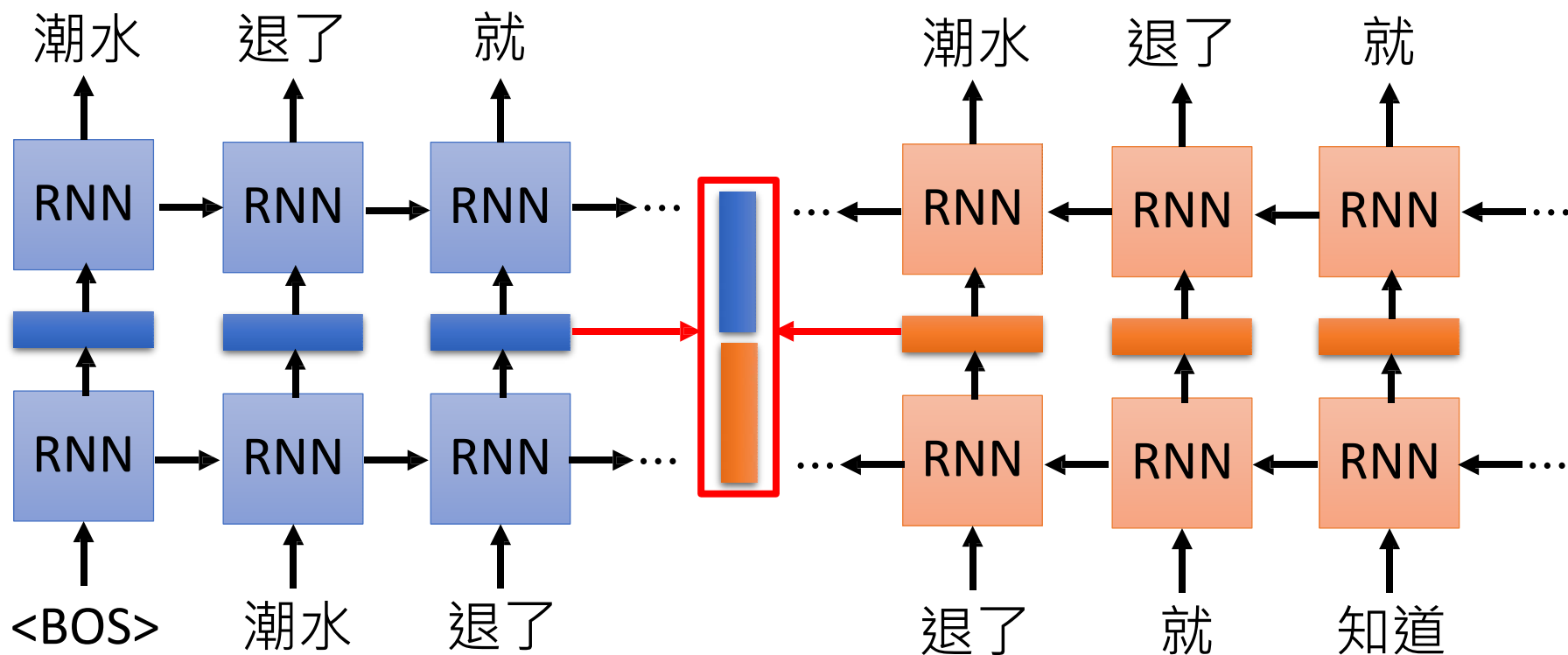
- Embeddings from Language Model (ELMO) (NAACL 2018 Best Paper)



$$\sum_{k=1}^N (\log p(t_k | t_1, \dots, t_{k-1}; \Theta_x, \vec{\Theta}_{LSTM}, \Theta_s) + \log p(t_k | t_{k+1}, \dots, t_N; \Theta_x, \overleftarrow{\Theta}_{LSTM}, \Theta_s)).$$

预训练任务

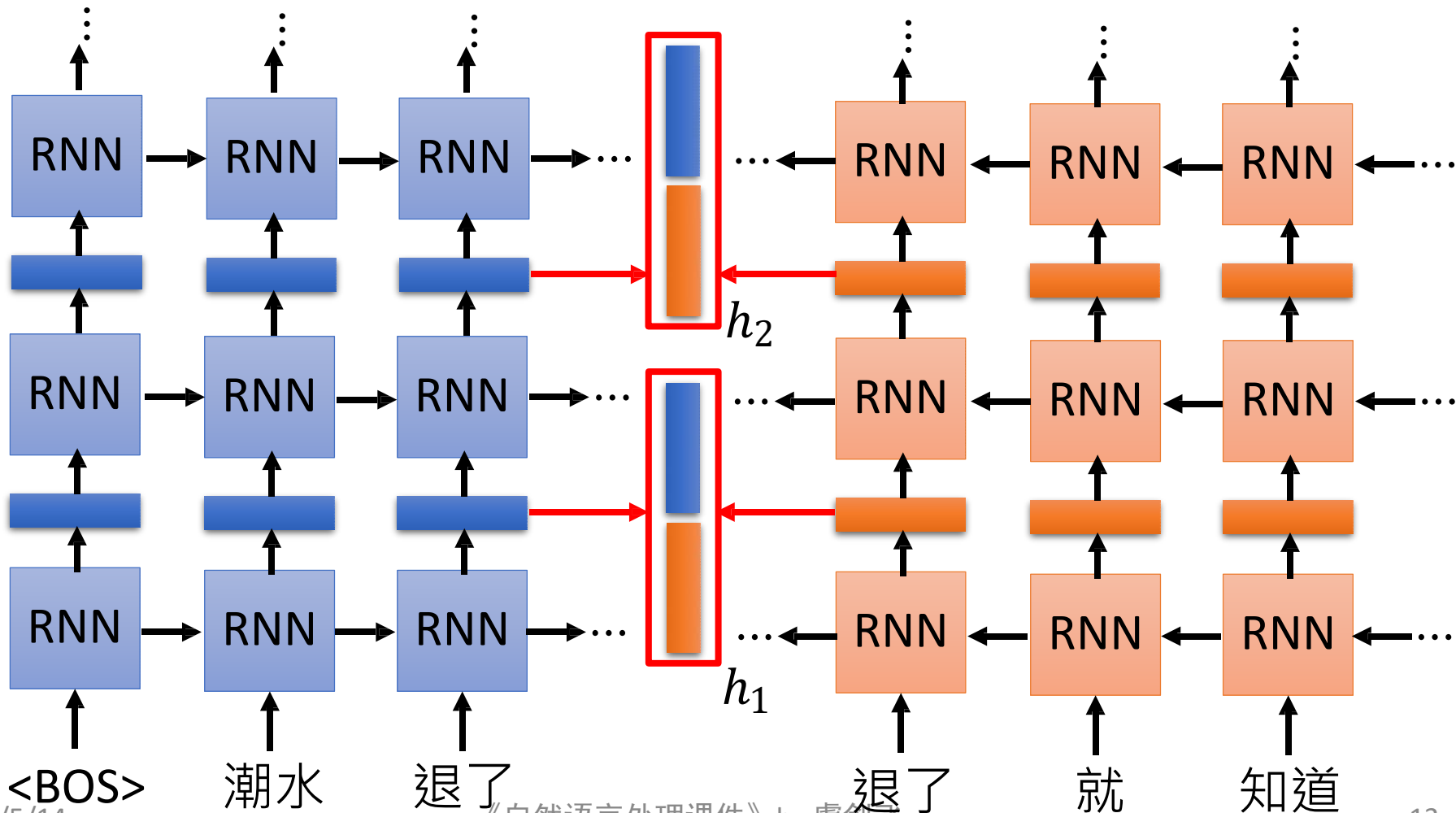
- Embeddings from Language Model (ELMO) (NAACL 2018 Best Paper)
 - RNN-based language models (trained from lots of sentences)
 - Bidirectional Modeling



预训练任务

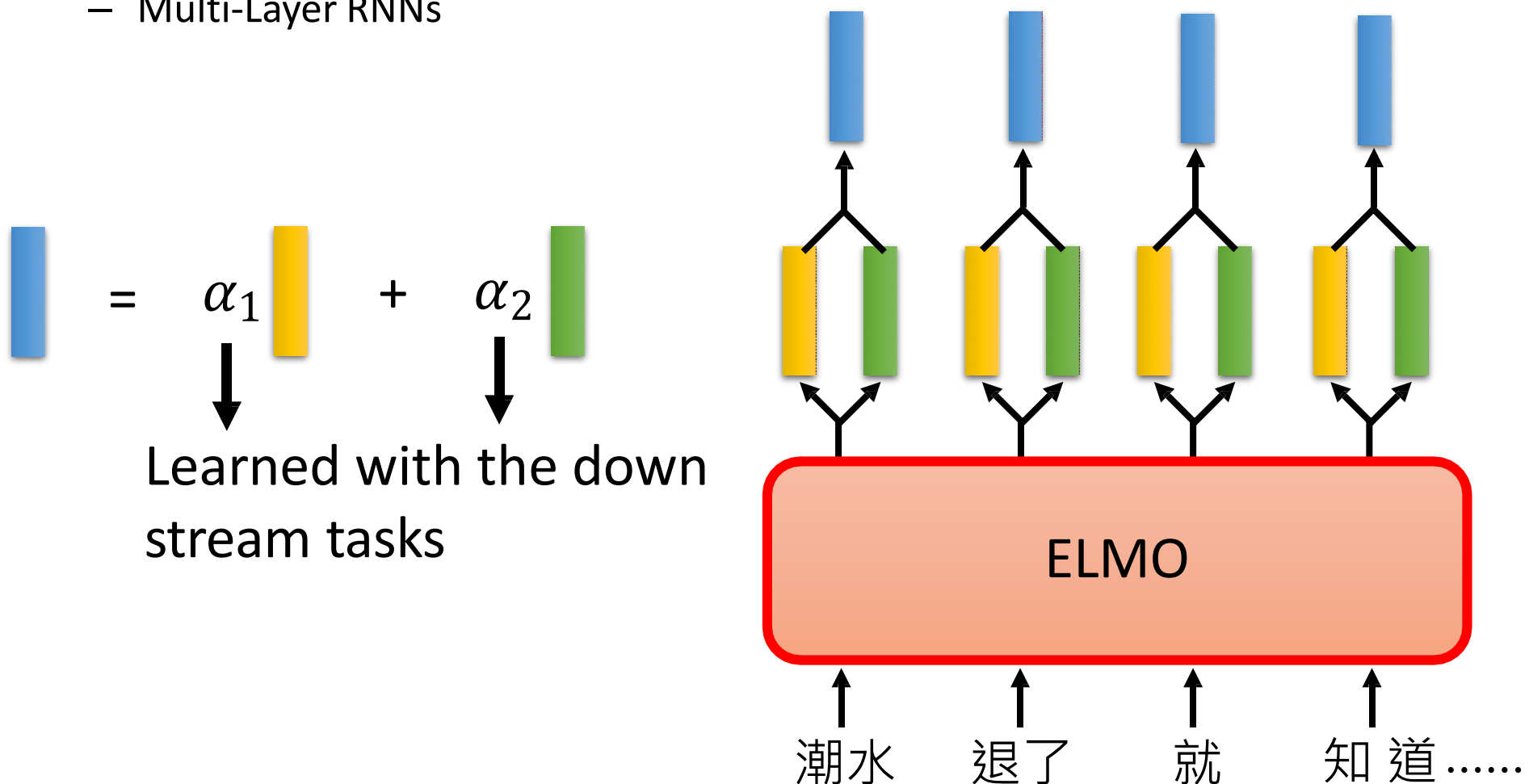
- Embeddings from Language Model (ELMO) (NAACL 2018 Best Paper)

– Multi-Layer RNNs



预训练任务

- Embeddings from Language Model (ELMO) (NAACL 2018 Best Paper)
 - Multi-Layer RNNs

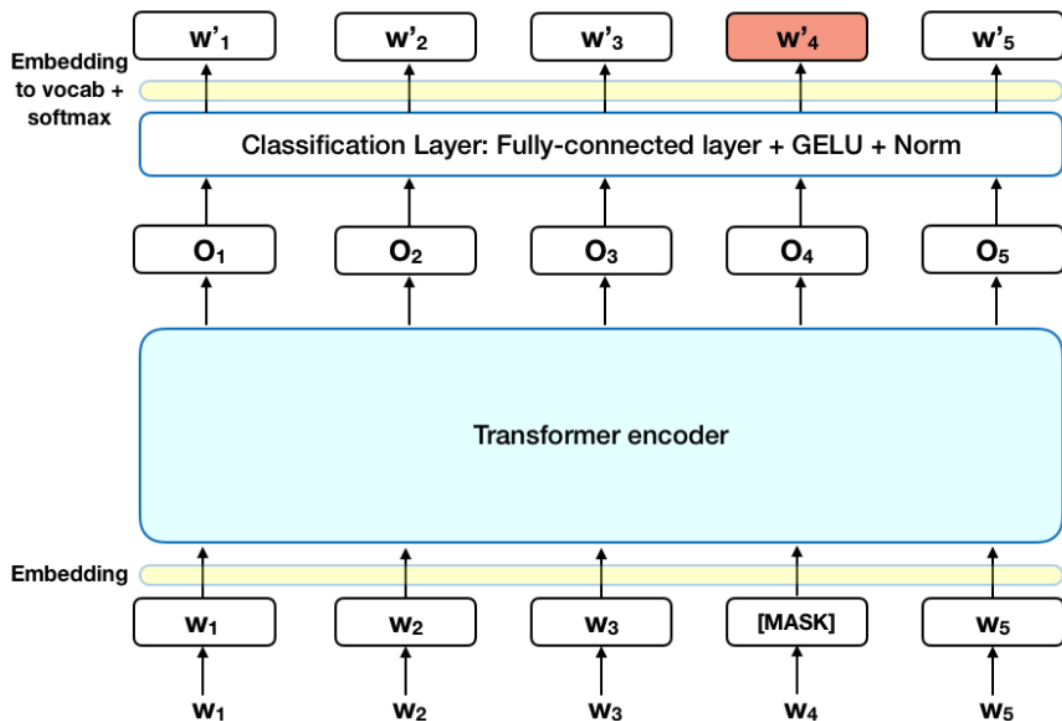


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预训练任务

- 自监督学习
 - Masked Language Modeling (MLM)

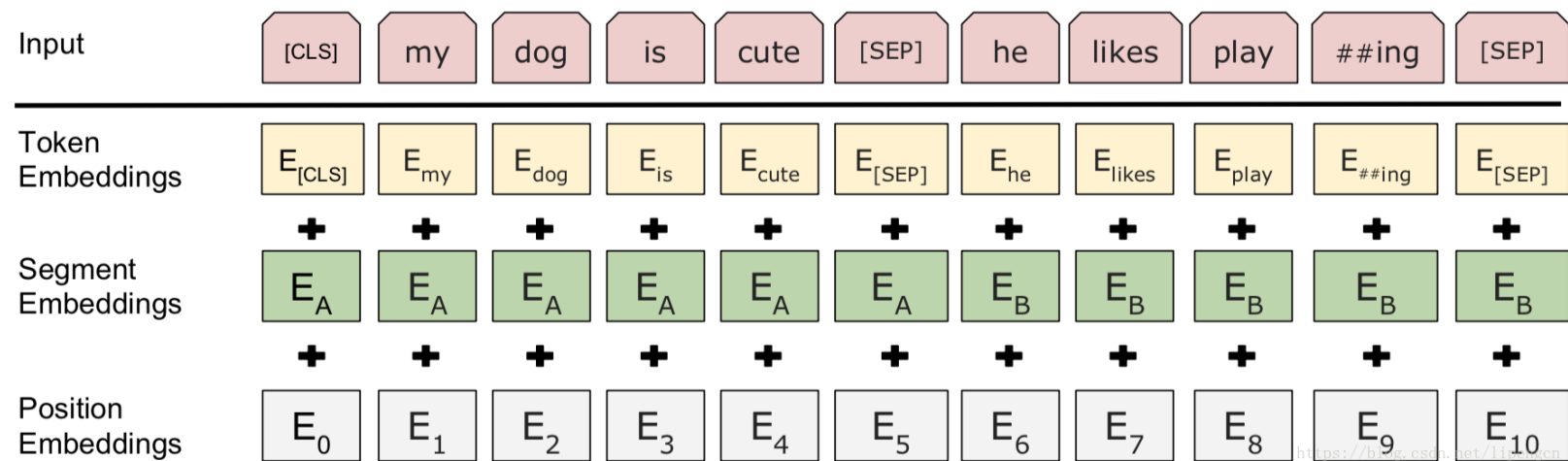


Bidirectional Encoder Representations from Transformers (BERT)

Devlin J, Chang M W, Lee K, et al. Bert: Pre-training of deep bidirectional transformers for language understanding. In NAACL, 2019. Best Paper Award

预训练任务

- 自监督学习
 - Next Sentence Prediction (NSP)



句子A

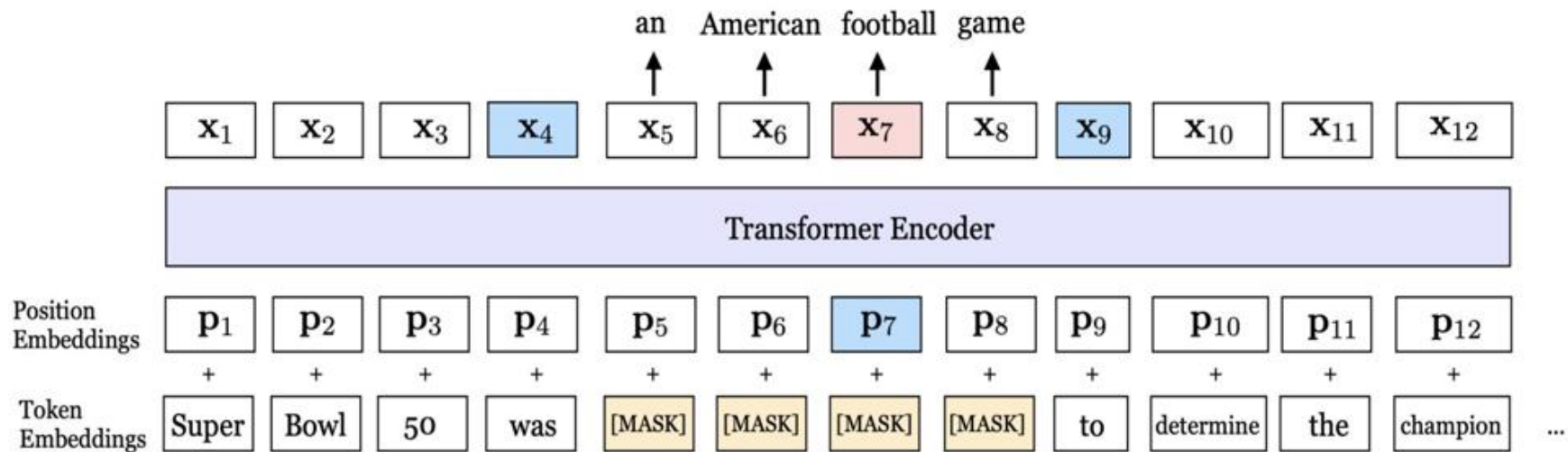


句子B

Devlin J, Chang M W, Lee K, et al. Bert: Pre-training of deep bidirectional transformers for language understanding. In NAACL, 2019. Best Paper Award

预训练任务

- 自监督学习
 - 预测一个范围内的所有词 (SpanBERT)



Joshi M, Chen D, Liu Y, et al. Spanbert: Improving pre-training by representing and predicting spans. Transactions of the Association for Computational Linguistics, 2020

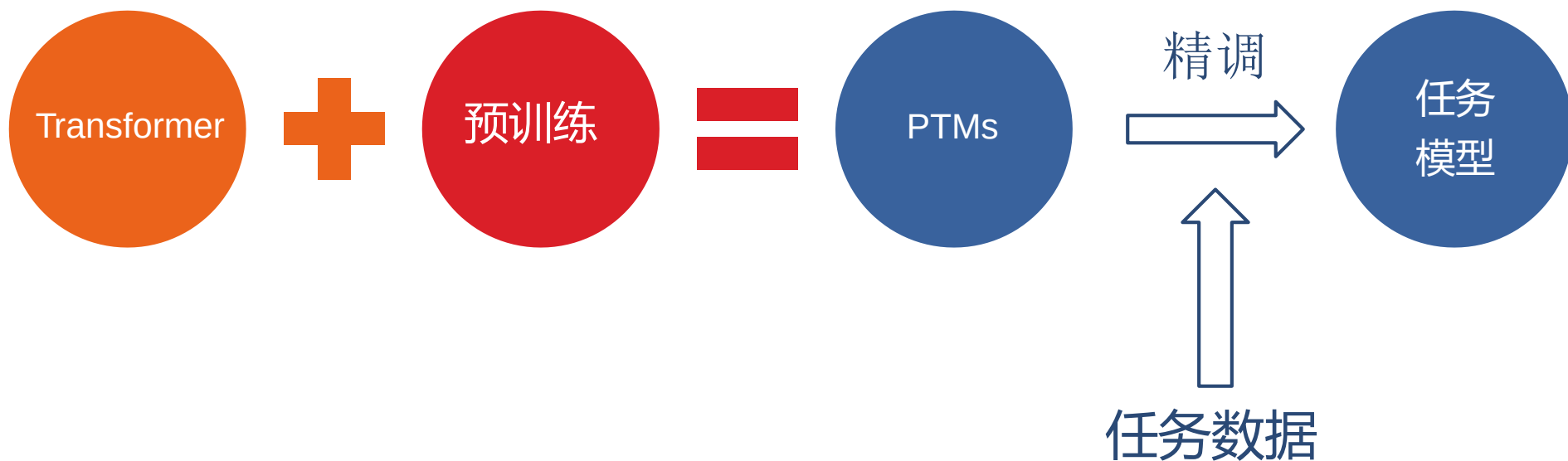
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下游任务

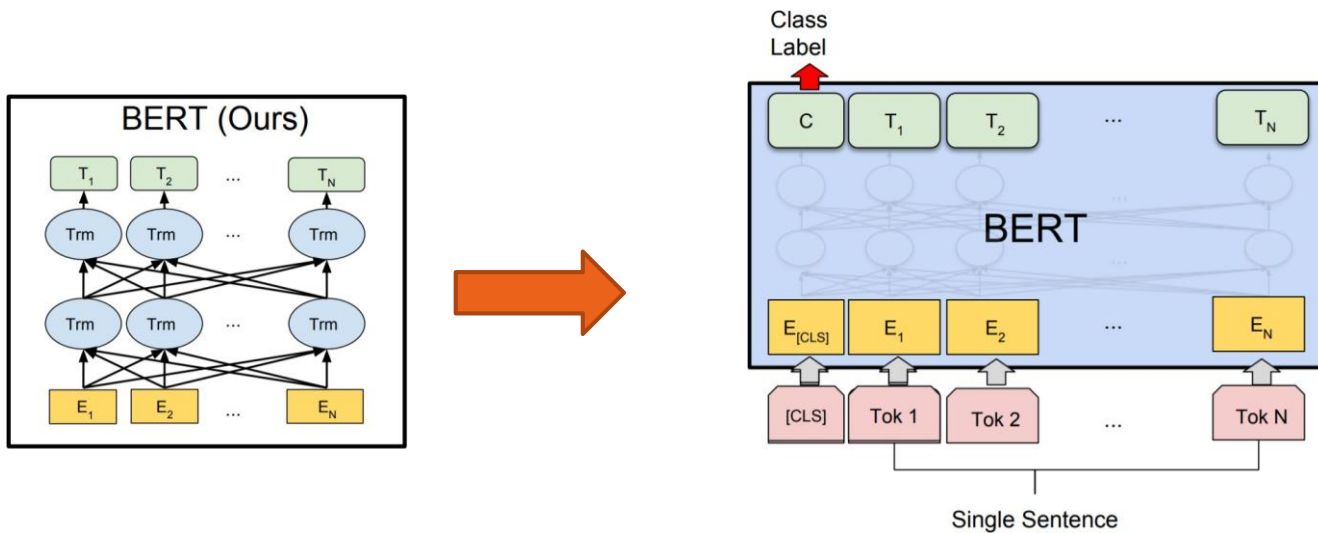
- 预训练+精调范式

迁移学习



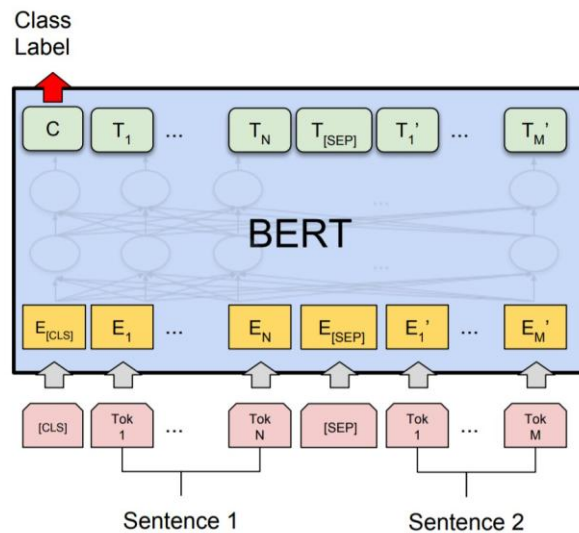
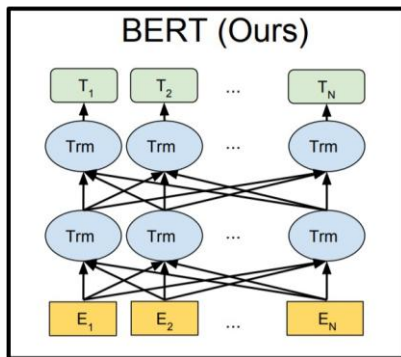
下游任务

- 预训练+精调范式
 - BERT



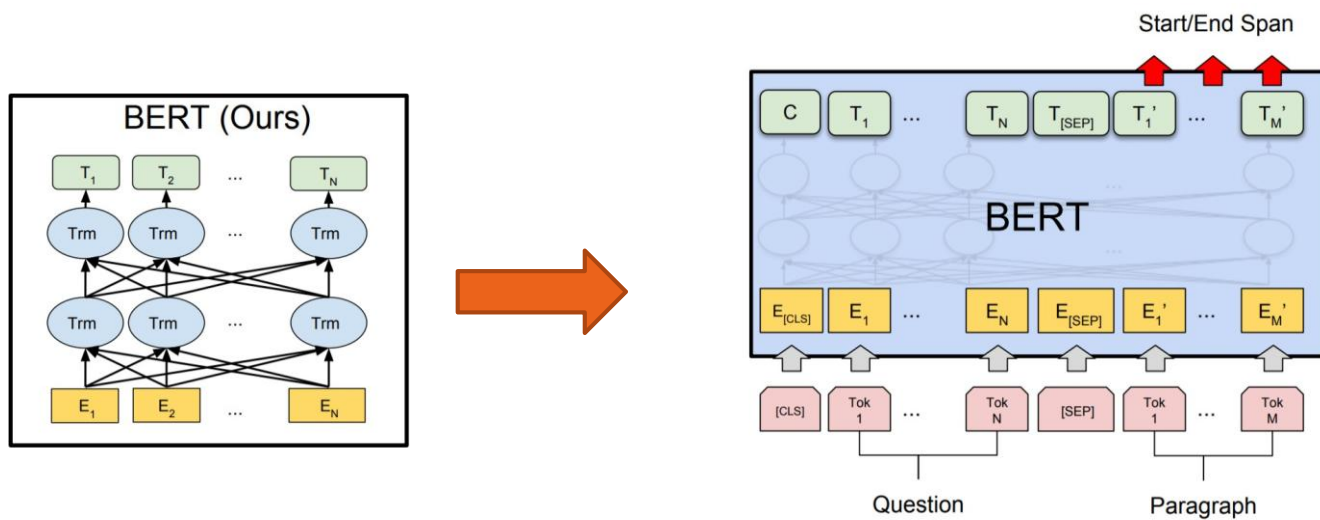
下游任务

- 预训练+精调范式
 - BERT



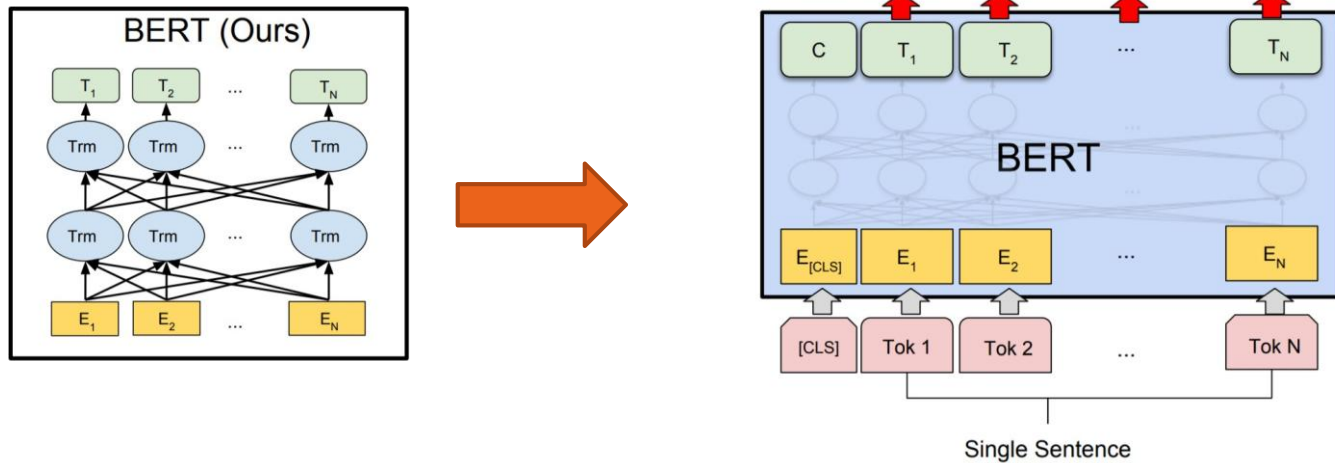
下游任务

- 预训练+精调范式
 - BERT



下游任务

- 预训练+精调范式
 - BERT



下游任务

- 预训练+精调范式
 - 预训练模型BERT的威力
 - General Language Understanding Evaluation (GLUE)
 - Corpus of Linguistic Acceptability (CoLA)
 - Stanford Sentiment Treebank (SST-2)
 - Microsoft Research Paraphrase Corpus (MRPC)
 - Quora Question Pairs (QQP)
 - Semantic Textual Similarity Benchmark (STS-B)
 - Multi-Genre Natural Language Inference (MNLI)
 - Question-answering NLI (QNLI)
 - Recognizing Textual Entailment (RTE)
 - Winograd NLI (WNLI)

<https://gluebenchmark.com/>

下游任务

- 预训练+精调范式
 - 预训练模型BERT的威力
 - BERT-base
 - 12 layers, $d_{\text{model}} = 768$, 12-headed attention
 - Total Parameters = $12 * 768 * 12 = 110\text{M}$
 - BERT-large
 - 24 layers, $d_{\text{model}} = 1,024$, 16-headed attention
 - Total Parameters = $24 * 1024 * 16 = 340\text{M}$

下游任务

- 预训练+精调范式
 - 预训练模型的威力

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

Table 1: GLUE Test results, scored by the evaluation server (<https://gluebenchmark.com/leaderboard>). The number below each task denotes the number of training examples. The “Average” column is slightly different than the official GLUE score, since we exclude the problematic WNLI set.⁸ BERT and OpenAI GPT are single-model, single task. F1 scores are reported for QQP and MRPC, Spearman correlations are reported for STS-B, and accuracy scores are reported for the other tasks. We exclude entries that use BERT as one of their components.

下游任务

- 预训练+精调范式
 - 预训练模型的威力

System	Dev		Test	
	EM	F1	EM	F1
Top Leaderboard Systems (Dec 10th, 2018)				
Human	-	-	82.3	91.2
#1 Ensemble - nlnet	-	-	86.0	91.7
#2 Ensemble - QANet	-	-	84.5	90.5
Published				
BiDAF+ELMo (Single)	-	85.6	-	85.8
R.M. Reader (Ensemble)	81.2	87.9	82.3	88.5
Ours				
BERT _{BASE} (Single)	80.8	88.5	-	-
BERT _{LARGE} (Single)	84.1	90.9	-	-
BERT _{LARGE} (Ensemble)	85.8	91.8	-	-
BERT _{LARGE} (Sgl.+TriviaQA)	84.2	91.1	85.1	91.8
BERT _{LARGE} (Ens.+TriviaQA)	86.2	92.2	87.4	93.2

Table 2: SQuAD 1.1 results. The BERT ensemble is 7x systems which use different pre-training checkpoints and fine-tuning seeds.

System	Dev		Test	
	EM	F1	EM	F1
Top Leaderboard Systems (Dec 10th, 2018)				
Human	86.3	89.0	86.9	89.5
#1 Single - MIR-MRC (F-Net)	-	-	74.8	78.0
#2 Single - nlnet	-	-	74.2	77.1
Published				
unet (Ensemble)	-	-	71.4	74.9
SLQA+ (Single)	-	-	71.4	74.4
Ours				
BERT _{LARGE} (Single)	78.7	81.9	80.0	83.1

Table 3: SQuAD 2.0 results. We exclude entries that use BERT as one of their components.

System	Dev	Test
ESIM+GloVe	51.9	52.7
ESIM+ELMo	59.1	59.2
OpenAI GPT	-	78.0
BERT _{BASE}	81.6	-
BERT _{LARGE}	86.6	86.3
Human (expert) [†]	-	85.0
Human (5 annotations) [†]	-	88.0

Table 4: SWAG Dev and Test accuracies. [†]Human performance is measured with 100 samples, as reported in the SWAG paper.

下游任务

- 预训练+精调范式
 - 预训练模型的威力

System	Dev F1	Test F1
ELMo (Peters et al., 2018a)	95.7	92.2
CVT (Clark et al., 2018)	-	92.6
CSE (Akbik et al., 2018)	-	93.1
Fine-tuning approach		
BERT _{LARGE}	96.6	92.8
BERT _{BASE}	96.4	92.4
Feature-based approach (BERT _{BASE})		
Embeddings	91.0	-
Second-to-Last Hidden	95.6	-
Last Hidden	94.9	-
Weighted Sum Last Four Hidden	95.9	-
Concat Last Four Hidden	96.1	-
Weighted Sum All 12 Layers	95.5	-

Table 7: CoNLL-2003 Named Entity Recognition results. Hyperparameters were selected using the Dev set. The reported Dev and Test scores are averaged over 5 random restarts using those hyperparameters.

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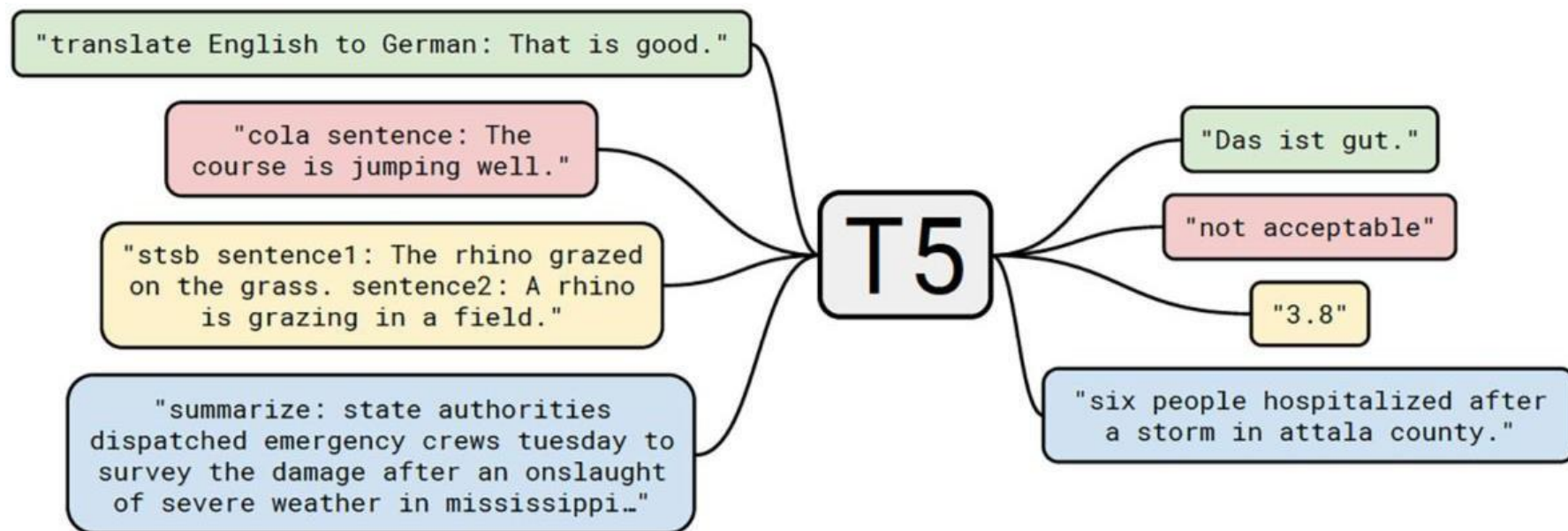
预训练任务

- 自监督学习
 - 典型任务

- Seq2Seq Language Modeling (LM)
- Seq2Seq Masked Language Modeling (MLM)
- Seq2Seq Permuted Language Modeling (PLM)
- Seq2Seq Masked Span Prediction
-

下游任务

- 统一建模成生成式任务
 - T5

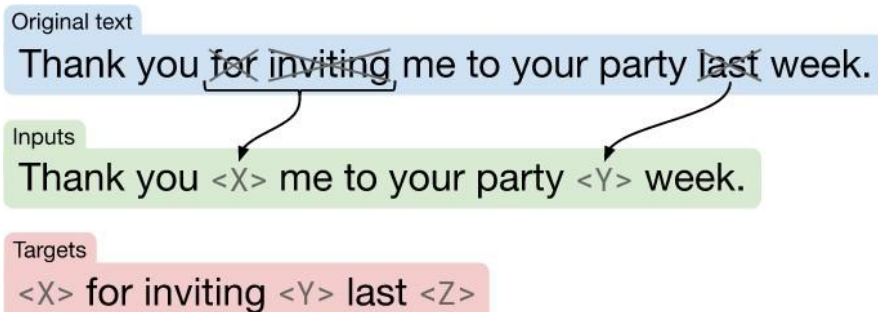


Text-to-Text Transfer Transformer (T5)

Raffel C, Shazeer N, Roberts A, et al. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. Journal of Machine Learning Research, 21, 1-67.

预训练任务

- T5
 - 自监督学习



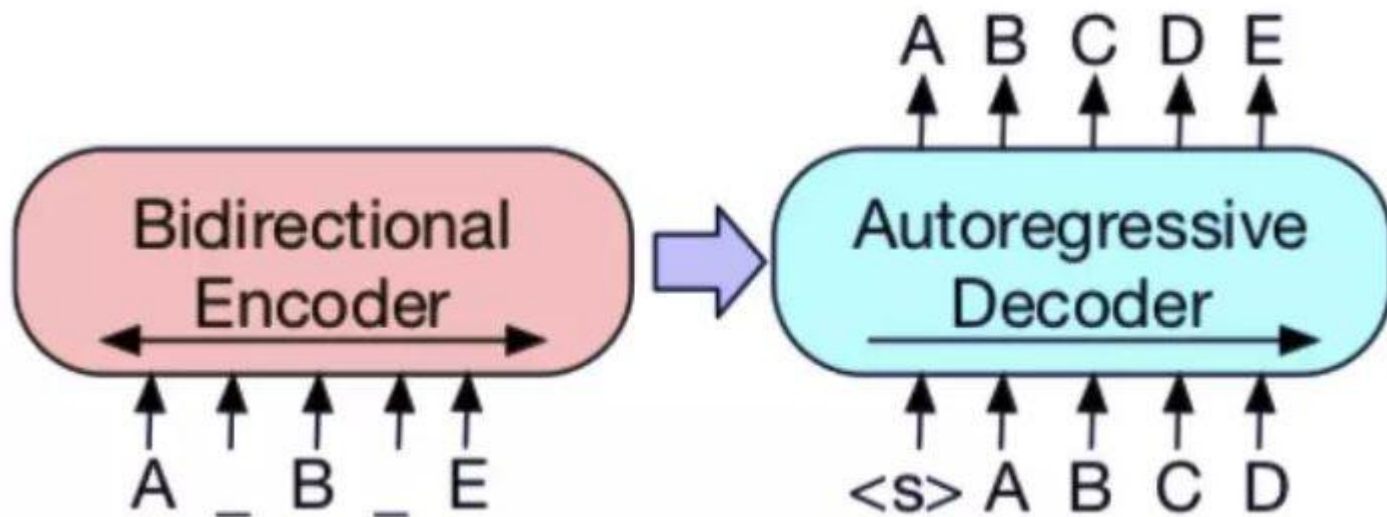
Objective	Inputs	Targets
Prefix language modeling	Thank you for inviting	me to your party last week .
BERT-style	Thank you <M> <M> me to your party apple week .	(original text)
Deshuffling	party me for your to . last fun you inviting week Thank	(original text)
I.i.d. noise, mask tokens	Thank you <M> <M> me to your party <M> week .	(original text)
I.i.d. noise, replace spans	Thank you <X> me to your party <Y> week .	<X> for inviting <Y> last <Z>
I.i.d. noise, drop tokens	Thank you me to your party week .	for inviting last
Random spans	Thank you <X> to <Y> week .	<X> for inviting me <Y> your party last <Z>

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预训练任务

- BART
 - 自监督学习
 - document corruption
 - reconstruction

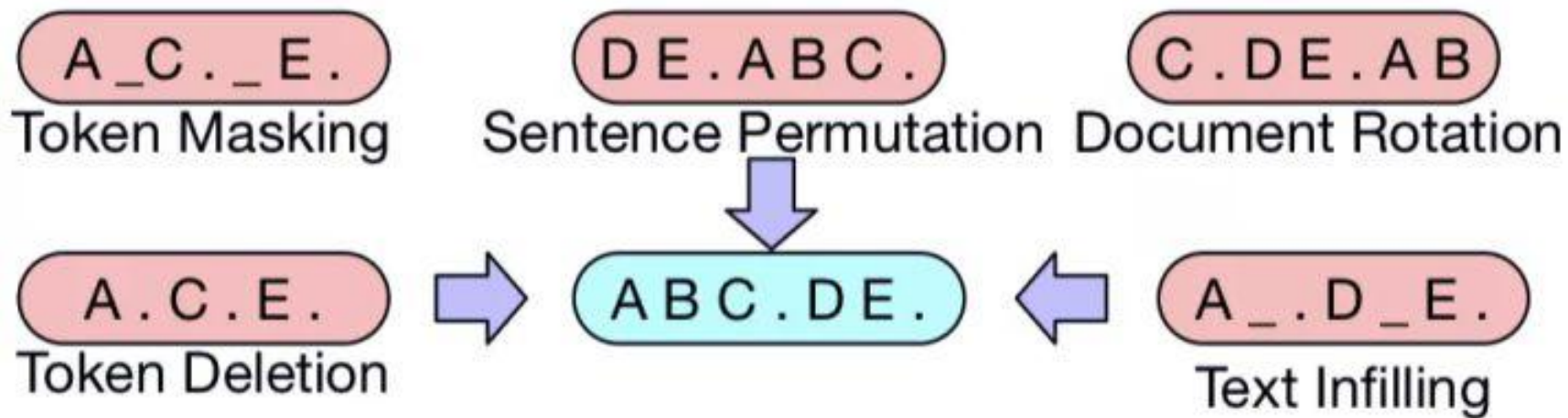


BART

Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., Stoyanov, V. and Zettlemoyer, L., 2020, July. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. In Proceedings of ACL (pp. 7871-7880).

预训练任务

- BART
 - 自监督学习
 - document corruption
 - reconstruction



Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., Stoyanov, V. and Zettlemoyer, L., 2020, July. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. In Proceedings of ACL (pp. 7871-7880).

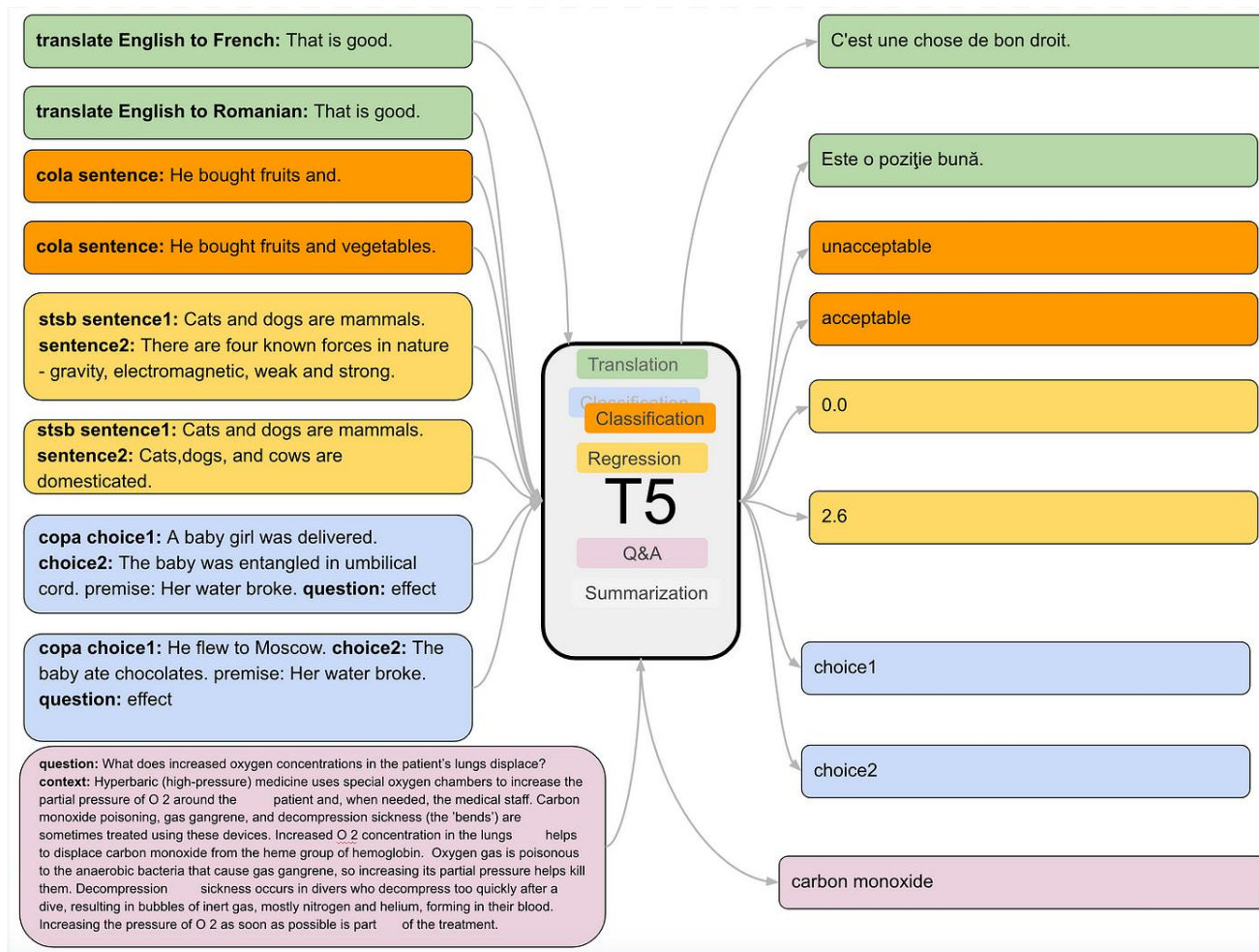
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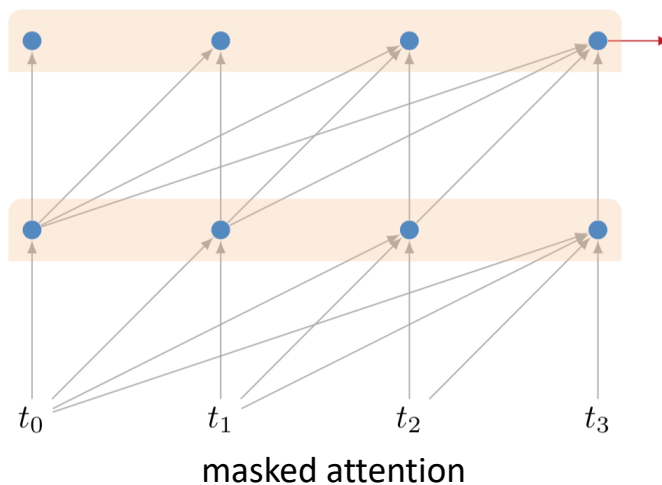
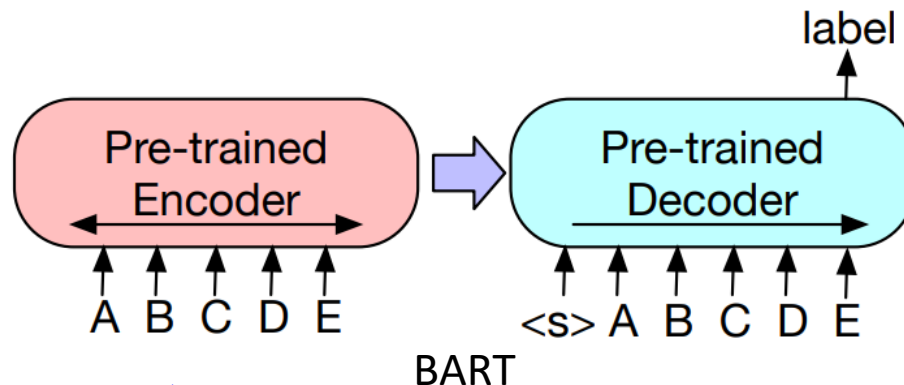
- 统一建模成生成式任务

– T5



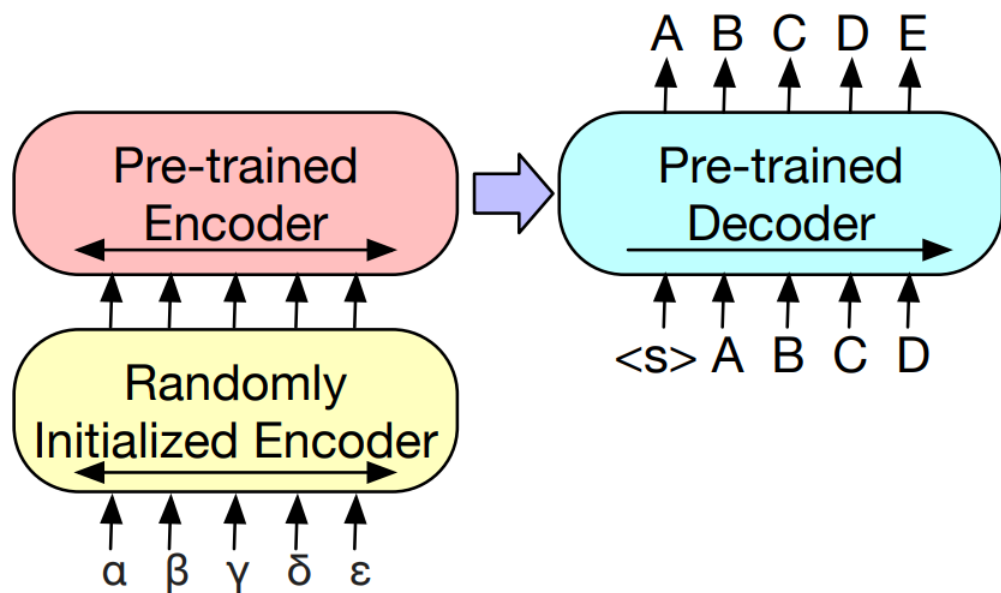
下游任务

- 统一建模生成式任务
 - BART for sequence classification



下游任务

- 统一建模成生成式任务
 - BART for machine translation



	RO-EN
Baseline	36.80
Fixed BART	36.29
Tuned BART	37.96

Table 8: BLEU scores of the baseline and BART on WMT'16 RO-EN augmented with back-translation data. BART improves over a strong back-translation baseline by using monolingual English pre-training.

下游任务

- 统一建模成生成式任务

- T5的威力

- T5-small: 6 layers, $d_{\text{model}} = 512$, $d_{\text{ff}} = 2,048$, 8-headed attention
 - T5-base: 12 layers, $d_{\text{model}} = 768$, $d_{\text{ff}} = 3,072$, 12-headed attention
 - T5-large: 24 layers, $d_{\text{model}} = 1,024$, $d_{\text{ff}} = 4,096$, 16-headed attention
 - T5-3B: 24 layers, $d_{\text{model}} = 1,024$, $d_{\text{ff}} = 16,384$, 32-headed attention
 - T5-11B: 24 layers, $d_{\text{model}} = 1,024$, $d_{\text{ff}} = 65,536$, 128-headed attention

Raffel C, Shazeer N, Roberts A, et al. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. Journal of Machine Learning Research, 21, 1-67.

下游任务

- 统一建模成生成式任务
 - T5的威力

Model	GLUE Average	CoLA Matthew's	SST-2 Accuracy	MRPC F1	MRPC Accuracy	STS-B Pearson	STS-B Spearman
Previous best	89.4 ^a	69.2 ^b	97.1 ^a	93.6^b	91.5^b	92.7 ^b	92.3 ^b
T5-Small	77.4	41.0	91.8	89.7	86.6	85.6	85.0
T5-Base	82.7	51.1	95.2	90.7	87.5	89.4	88.6
T5-Large	86.4	61.2	96.3	92.4	89.9	89.9	89.2
T5-3B	88.5	67.1	97.4	92.5	90.0	90.6	89.8
T5-11B	90.3	71.6	97.5	92.8	90.4	93.1	92.8

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下游任务

- 统一建模成生成式任务
 - T5的威力

Model	QQP F1	QQP Accuracy	MNLI-m Accuracy	MNLI-mm Accuracy	QNLI Accuracy	RTE Accuracy	WNLI Accuracy
Previous best	74.8 ^c	90.7^b	91.3 ^a	91.0 ^a	99.2^a	89.2 ^a	91.8 ^a
T5-Small	70.0	88.0	82.4	82.3	90.3	69.9	69.2
T5-Base	72.6	89.4	87.1	86.2	93.7	80.1	78.8
T5-Large	73.9	89.9	89.9	89.6	94.8	87.2	85.6
T5-3B	74.4	89.7	91.4	91.2	96.3	91.1	89.7
T5-11B	75.1	90.6	92.2	91.9	96.9	92.8	94.5

Raffel C, Shazeer N, Roberts A, et al. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. Journal of Machine Learning Research, 21, 1-67.

下游任务

- 统一建模成生成式任务
 - T5的威力

Model	SQuAD EM	SQuAD F1	SuperGLUE Average	BoolQ Accuracy	CB F1	CB Accuracy	COPA Accuracy
Previous best	90.1 ^a	95.5 ^a	84.6 ^d	87.1 ^d	90.5 ^d	95.2 ^d	90.6 ^d
T5-Small	79.10	87.24	63.3	76.4	56.9	81.6	46.0
T5-Base	85.44	92.08	76.2	81.4	86.2	94.0	71.2
T5-Large	86.66	93.79	82.3	85.4	91.6	94.8	83.4
T5-3B	88.53	94.95	86.4	89.9	90.3	94.4	92.0
T5-11B	91.26	96.22	88.9	91.2	93.9	96.8	94.8

Raffel C, Shazeer N, Roberts A, et al. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. Journal of Machine Learning Research, 21, 1-67.

下游任务

- 统一建模成生成式任务
 - T5的威力

Model	MultiRC F1a	MultiRC EM	ReCoRD F1	ReCoRD Accuracy	RTE Accuracy	WiC Accuracy	WSC Accuracy
Previous best	84.4 ^d	52.5 ^d	90.6 ^d	90.0 ^d	88.2 ^d	69.9 ^d	89.0 ^d
T5-Small	69.3	26.3	56.3	55.4	73.3	66.9	70.5
T5-Base	79.7	43.1	75.0	74.2	81.5	68.3	80.8
T5-Large	83.3	50.7	86.8	85.9	87.8	69.3	86.3
T5-3B	86.8	58.3	91.2	90.4	90.7	72.1	90.4
T5-11B	88.1	63.3	94.1	93.4	92.5	76.9	93.8

Model	WMT EnDe BLEU	WMT EnFr BLEU	WMT EnRo BLEU	CNN/DM ROUGE-1	CNN/DM ROUGE-2	CNN/DM ROUGE-L
Previous best	33.8^e	43.8^e	38.5^f	43.47 ^g	20.30 ^g	40.63 ^g
T5-Small	26.7	36.0	26.8	41.12	19.56	38.35
T5-Base	30.9	41.2	28.0	42.05	20.34	39.40
T5-Large	32.0	41.5	28.1	42.50	20.68	39.75
T5-3B	31.8	42.6	28.2	42.72	21.02	39.94
T5-11B	32.1	43.4	28.1	43.52	21.55	40.69

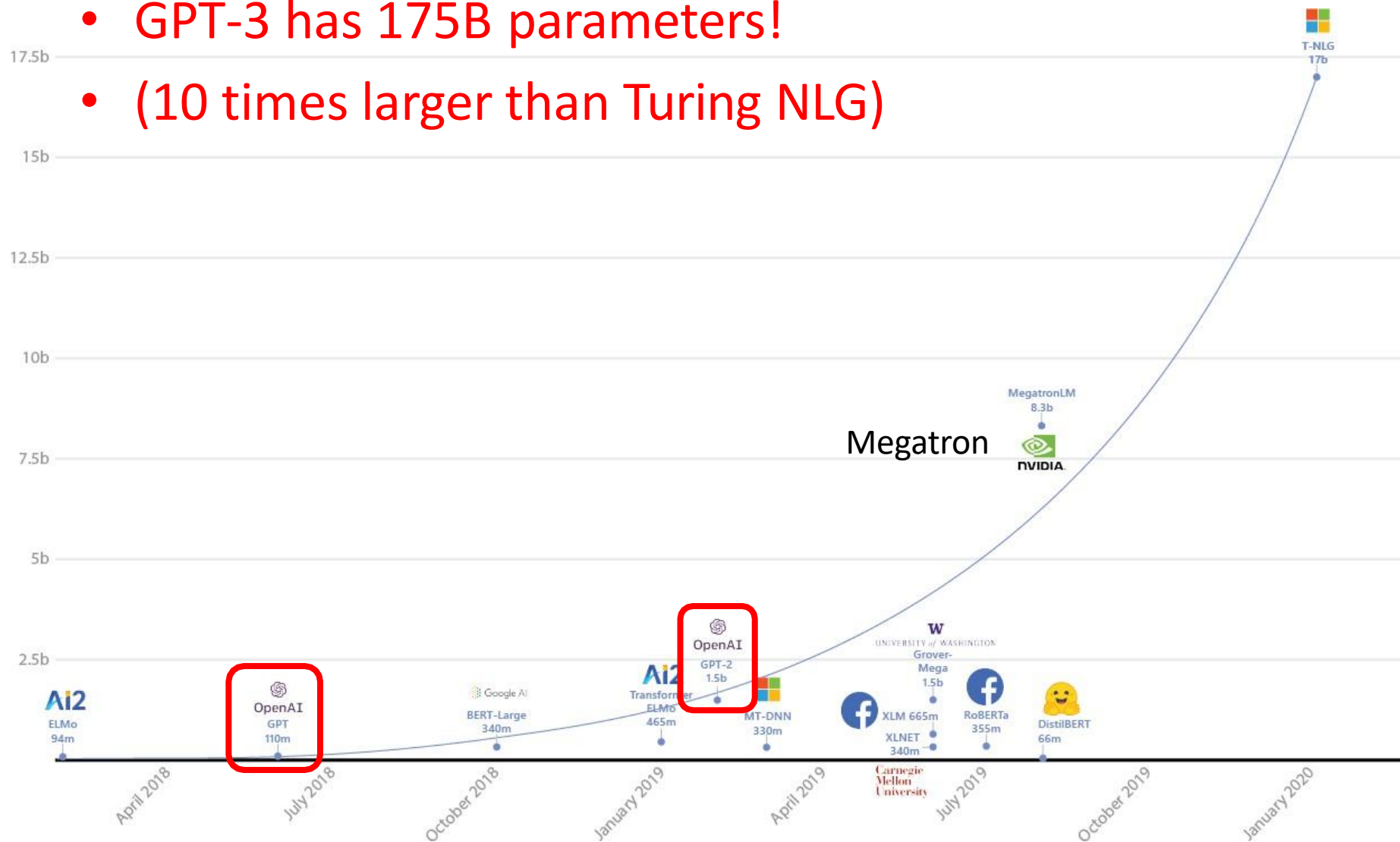
Raffel C, Shazeer N, Roberts A, et al. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. Journal of Machine Learning Research, 21, 1-67.

本章内容

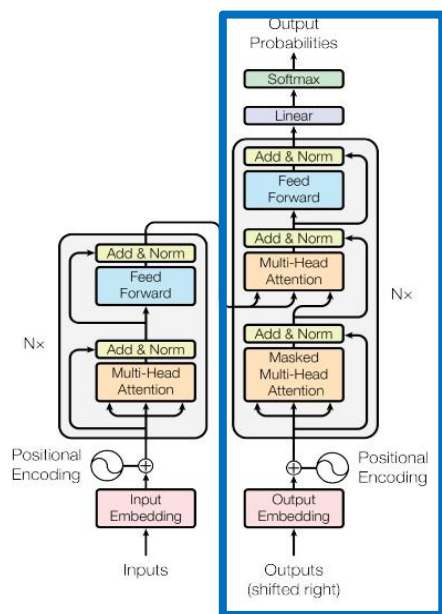
- 8.1 引言
- 8.2 自注意力机制
- 8.3 Transformer网络架构
- 8.4 预训练模型及其应用
 - 8.4.1 基于Transformer编码器的预训练语言模型
 - 8.4.2 基于Transformer的预训练Seq2Seq模型
 - 8.4.3 基于Transformer解码器的预训练语言模型
 - 预训练任务
 - 下游任务
 - 8.4.4 基于Transformer的文本-视觉预训练模型
- 8.5 本章小结

预训练任务

- GPT-3 has 175B parameters!
- (10 times larger than Turing NLG)

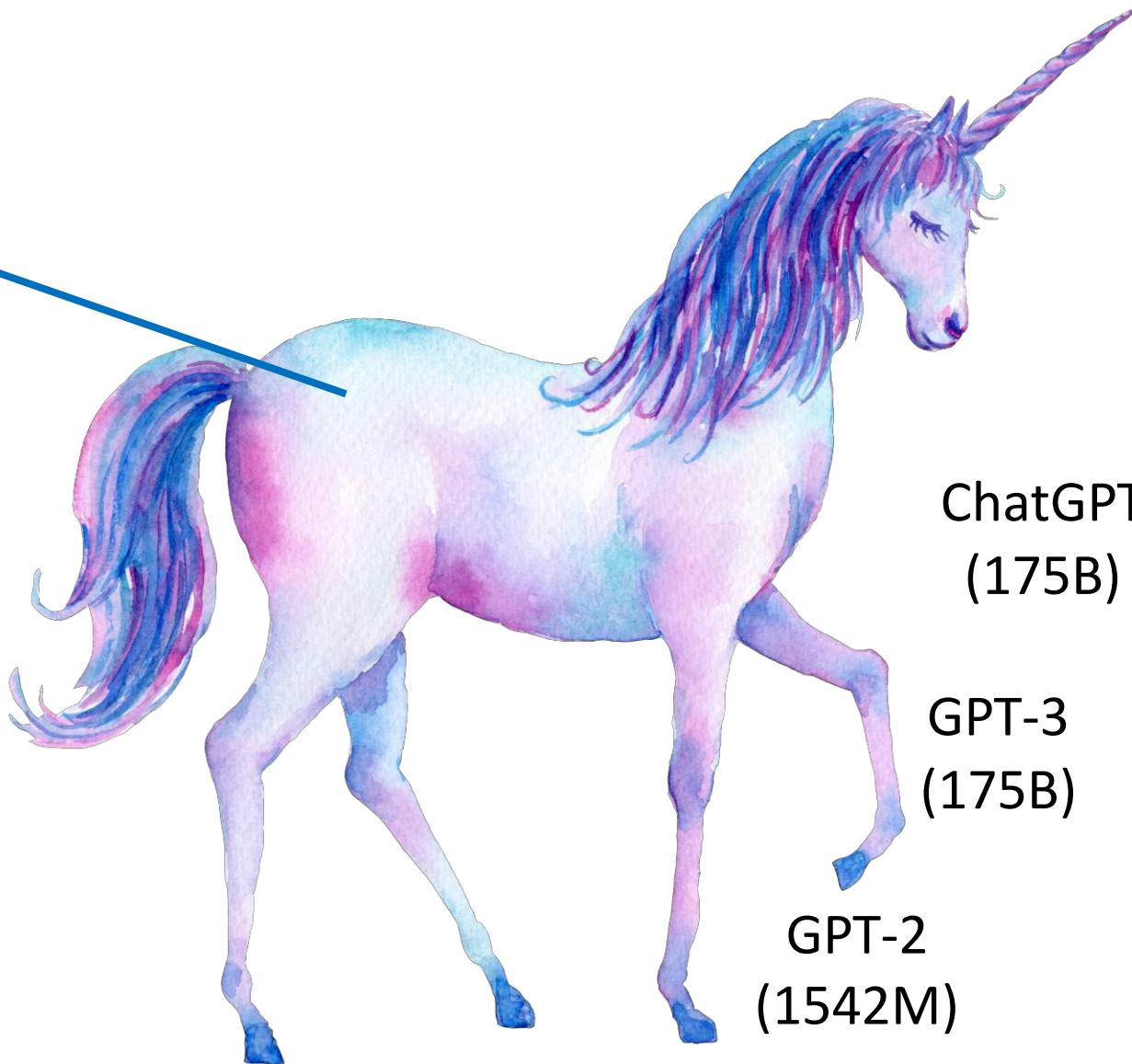


预训练任务



BERT
(340M)

ELMO
(94M)



ChatGPT
(175B)

GPT-3
(175B)

GPT-2
(1542M)

预训练任务

Improving Language Understanding by Generative Pre-Training

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Abstract

Natural language understanding comprises a wide range of diverse tasks such as textual entailment, question answering, semantic similarity assessment, and document classification. Although large unlabeled text corpora are abundant, labeled data for learning these specific tasks is scarce, making it challenging for discriminatively trained models to perform adequately. We demonstrate that large gains on these tasks can be realized by *generative pre-training* of a language model on a diverse corpus of unlabeled text, followed by *discriminative fine-tuning* on each specific task. In contrast to previous approaches, we make use of task-aware input transformations during fine-tuning to achieve effective transfer while requiring minimal changes to the model architecture. We demonstrate the effectiveness of our approach on a wide range of benchmarks for natural language understanding. Our general task-agnostic model outperforms discriminatively trained models that use architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test), 5.7% on question answering (RACE), and 1.5% on textual entailment (MultiNLI).

1 Introduction

The ability to learn effectively from raw text is crucial to alleviating the dependence on supervised learning in natural language processing (NLP). Most deep learning methods require substantial amounts of manually labeled data, which restricts their applicability in many domains that suffer from a dearth of annotated resources [61]. In these situations, models that can leverage linguistic information from unlabeled data provide a valuable alternative to gathering more annotation, which can be time-consuming and expensive. Further, even in cases where considerable supervision is available, learning good representations in an unsupervised fashion can provide a significant performance boost. The most compelling evidence for this so far has been the extensive use of pre-trained word embeddings [10, 39, 42] to improve performance on a range of NLP tasks [8, 11, 26, 45].

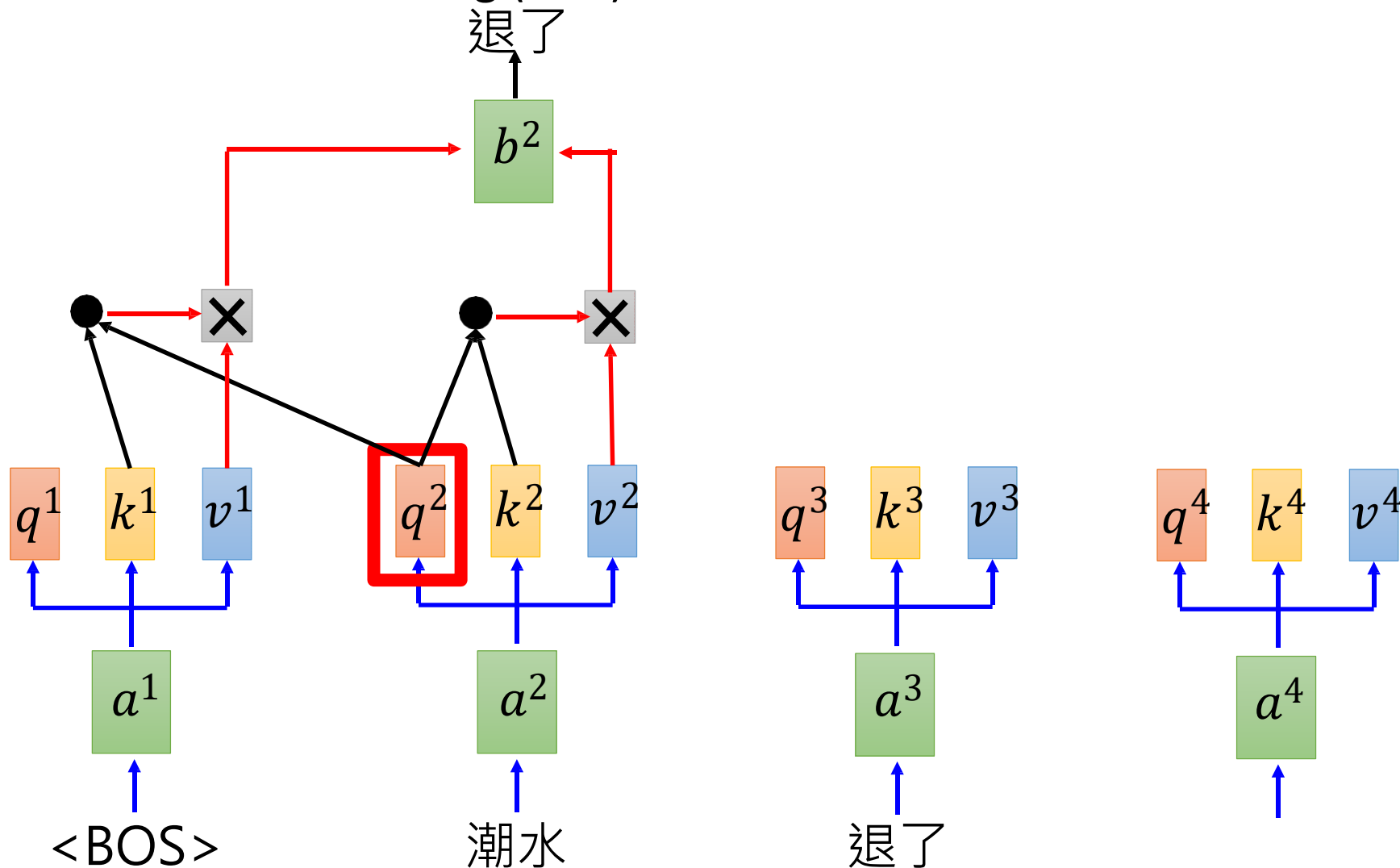
Leveraging more than word-level information from unlabeled text, however, is challenging for two main reasons. First, it is unclear what type of optimization objectives are most effective at learning text representations that are useful for transfer. Recent research has looked at various objectives such as language modeling [44], machine translation [38], and discourse coherence [22], with each method outperforming the others on different tasks.¹ Second, there is no consensus on the most effective way to transfer these learned representations to the target task. Existing techniques involve a combination of making task-specific changes to the model architecture [43, 44], using intricate learning schemes [21] and adding auxiliary learning objectives [50]. These uncertainties have made it difficult to develop effective semi-supervised learning approaches for language processing.

¹<https://gluebenchmark.com/leaderboard>

Preprint. Work in progress.

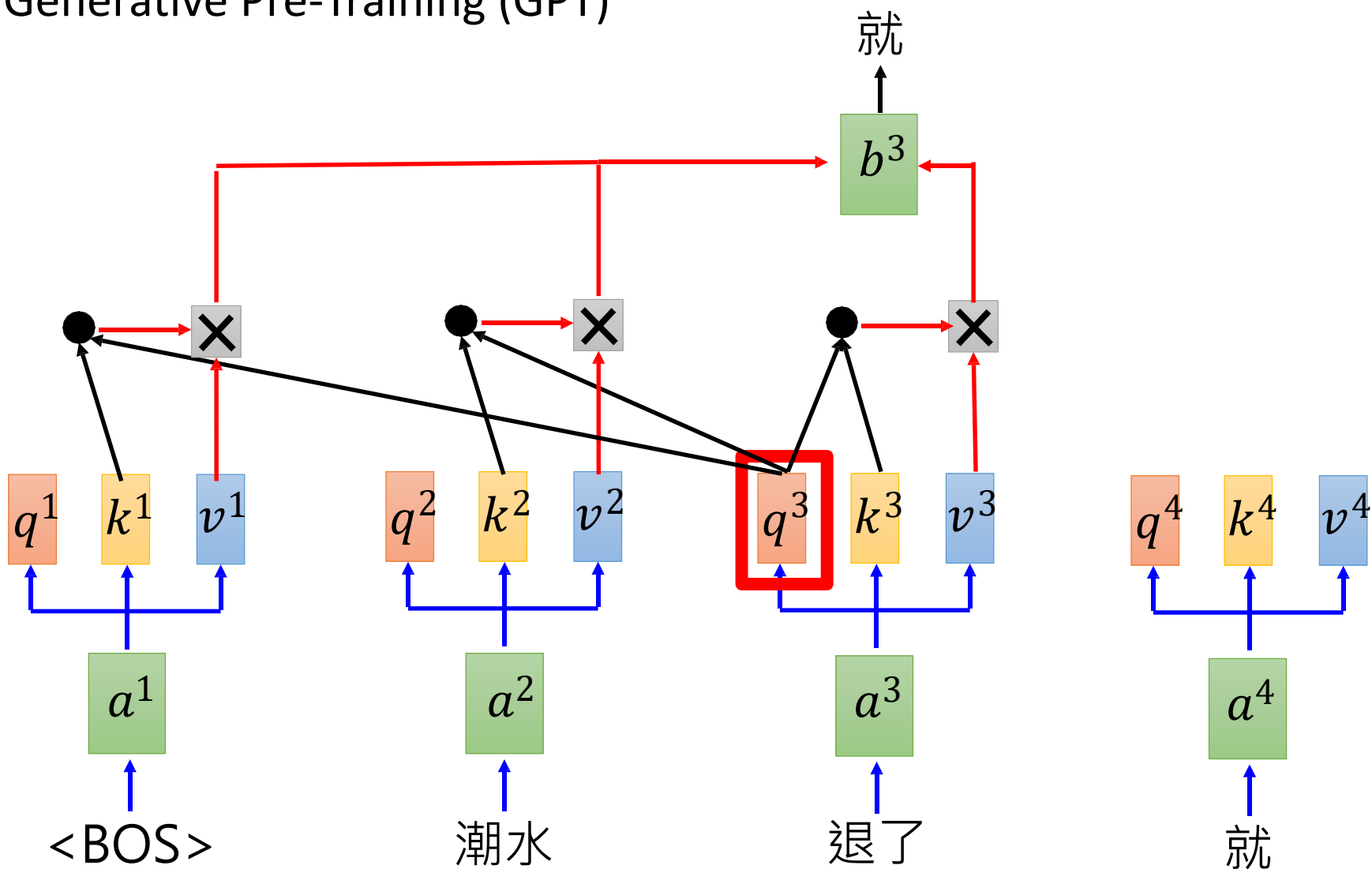
预训练任务

- Generative Pre-Training (GPT)



预训练任务

- Generative Pre-Training (GPT)

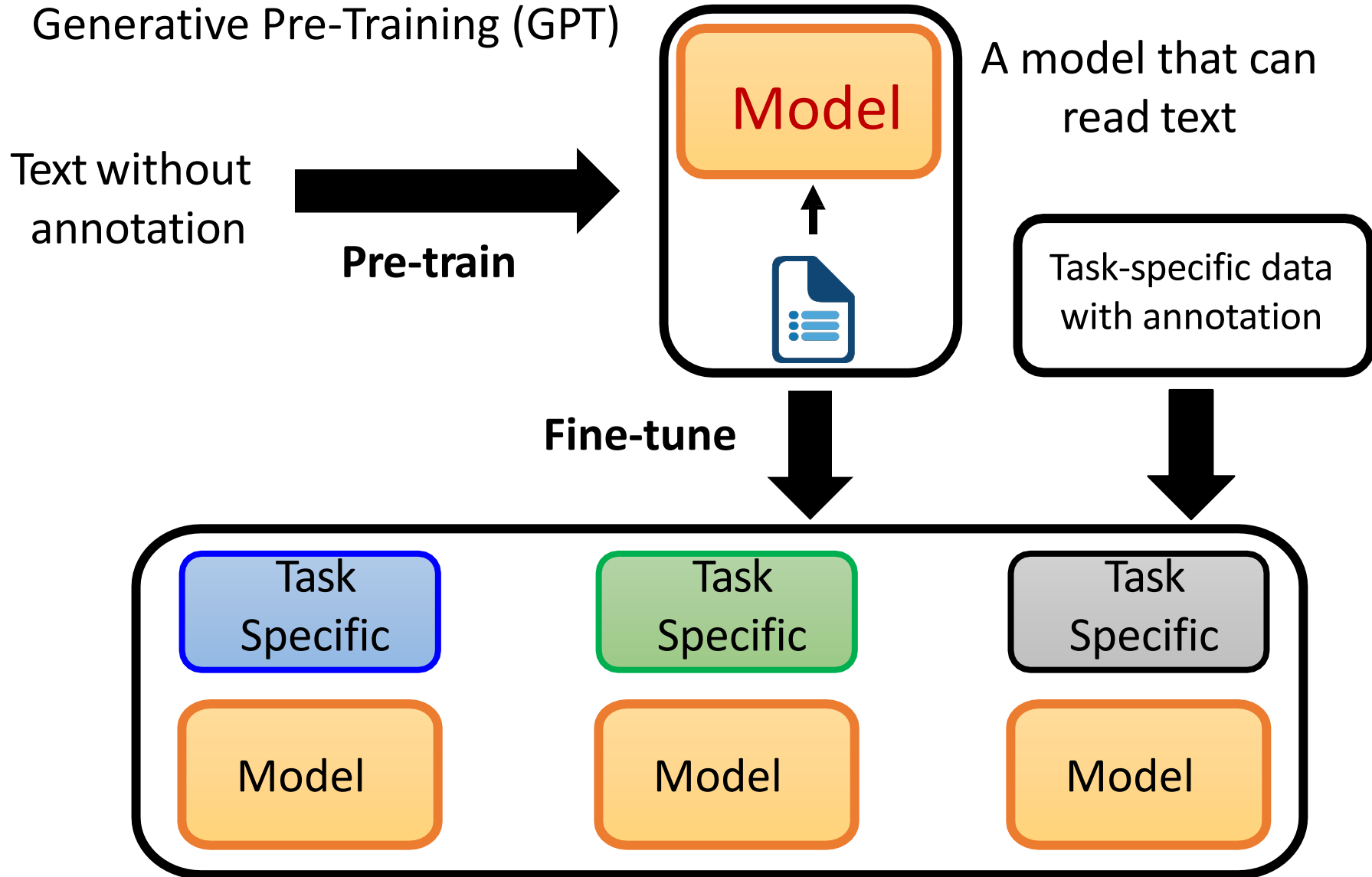


本章内容

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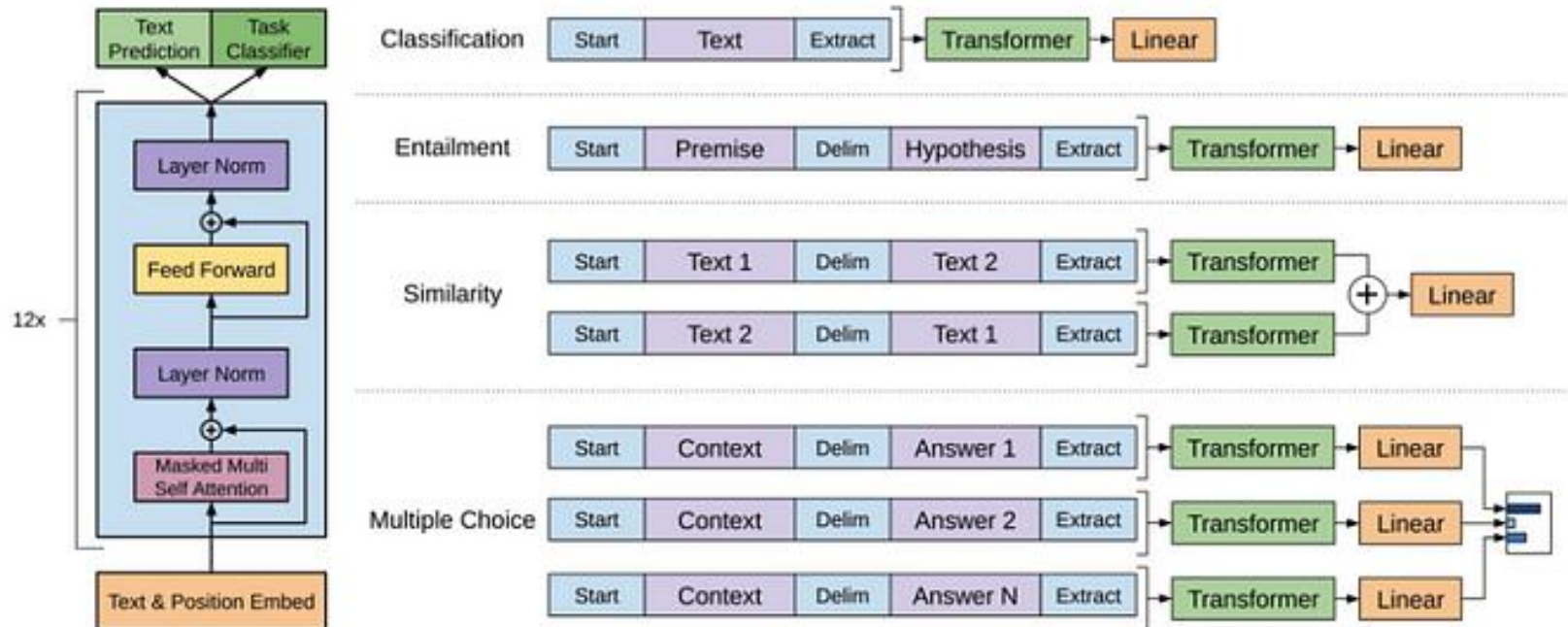
下游任务

- Generative Pre-Training (GPT)



下游任务

- Generative Pre-Training (GPT)



下游任务

- Generative Pre-Training 3 (GPT-3)

Language Models are Few-Shot Learners

Tom B. Brown*	Benjamin Mann*	Nick Ryder*	Melanie Subbiah*
Jared Kaplan†	Prafulla Dhariwal	Arvind Neelakantan	Pranav Shyam
Girish Sastry	Amanda Askell	Sandhini Agarwal	Ariel Herbert-Voss
Gretchen Krueger	Tom Henighan	Rewon Child	Aditya Ramesh
Daniel M. Ziegler	Jeffrey Wu	Clemens Winter	
Christopher Hesse	Mark Chen	Eric Sigler	Mateusz Litwin
Scott Gray			
Benjamin Chess	Jack Clark	Christopher Berner	
Sam McCandlish	Alec Radford	Ilya Sutskever	Dario Amodei

Abstract

We demonstrate that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even becoming competitive with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks. We also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora.

1 Introduction

NLP has shifted from learning task-specific representations and designing task-specific architectures to using task-agnostic pre-training and task-agnostic architectures. This shift has led to substantial progress on many challenging NLP tasks such as reading comprehension, question answering, textual entailment, among others. Even though the architecture and initial representations are now task-agnostic, a final task-specific step remains: fine-tuning on a large dataset of examples to adapt a task-agnostic model to perform a desired task.

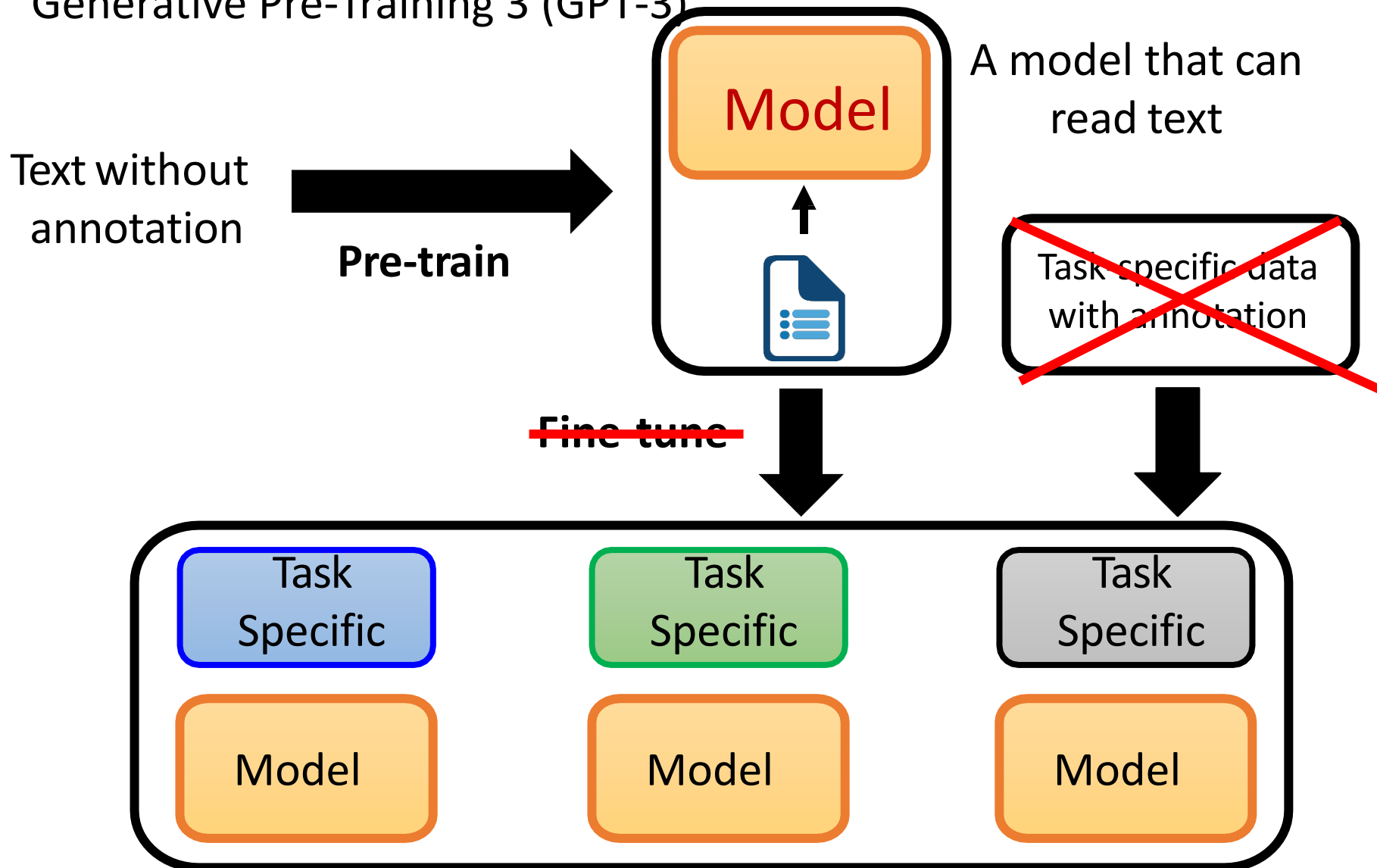
Recent work [RWC⁺19] suggested this final step may not be necessary. [RWC⁺19] demonstrated that a single pretrained language model can be zero-shot transferred to perform standard NLP tasks

*Equal contribution

†Johns Hopkins University, OpenAI

下游任务

- Generative Pre-Training 3 (GPT-3)



下游任务

- 上下文学习（零样本/小样本学习）
 - “In-context” Learning

Few-shot Learning (no gradient descent)

1	Translate English to French:	← task description
2	sea otter => loutre de mer	← examples
3	peppermint => menthe poivrée	←
4	plush girafe => girafe peluche	←
5	cheese =>	← prompt

One-shot Learning

1	Translate English to French:	← task description
2	sea otter => loutre de mer	← example
3	cheese =>	← prompt

Zero-shot Learning

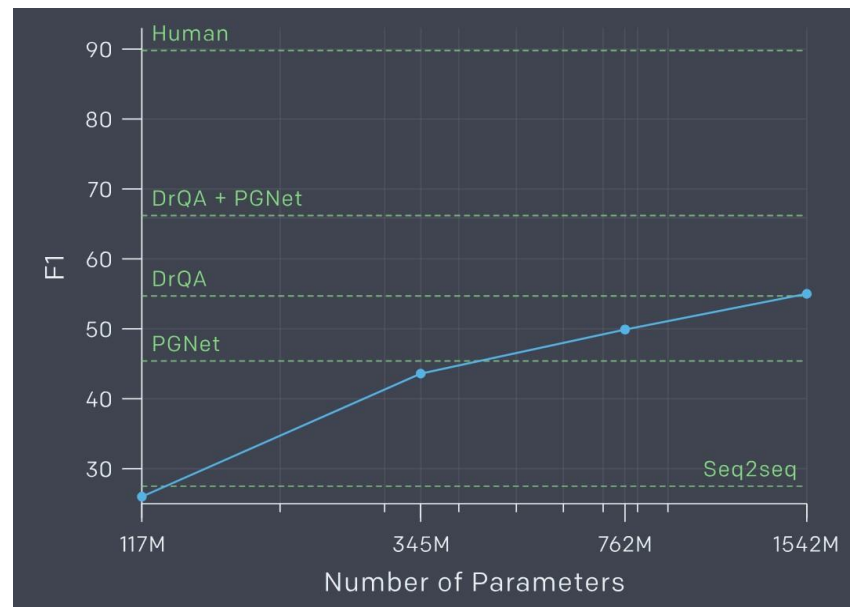
1	Translate English to French:	← task description
2	cheese =>	← prompt

下游任务

- 上下文学习（零样本/小样本学习）
 - “In-context” Learning

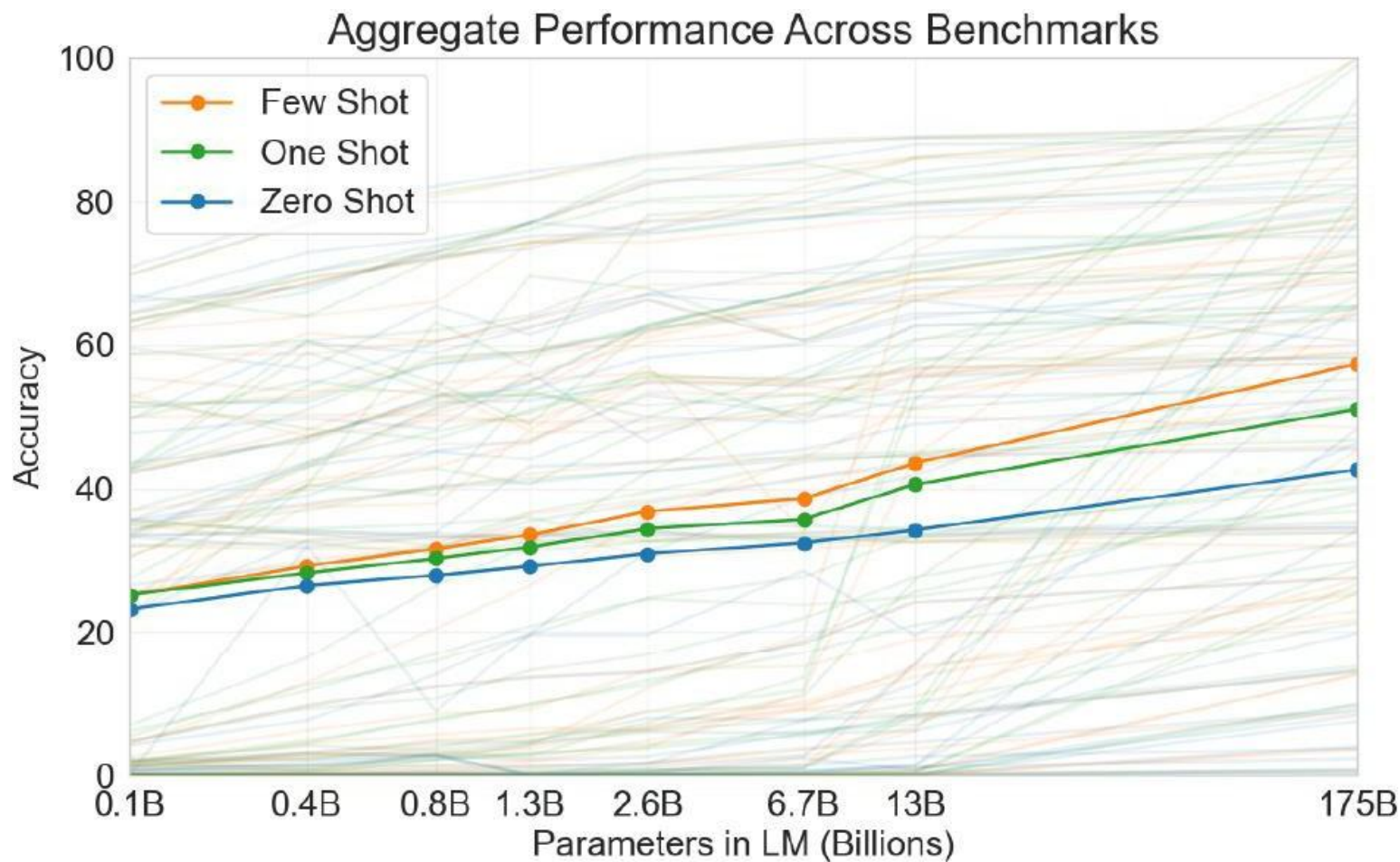
- Reading Comprehension**

$d_1, d_2, \dots, d_N,$
“Q:”, $q_1, q_2, \dots, q_N,$
“A:”



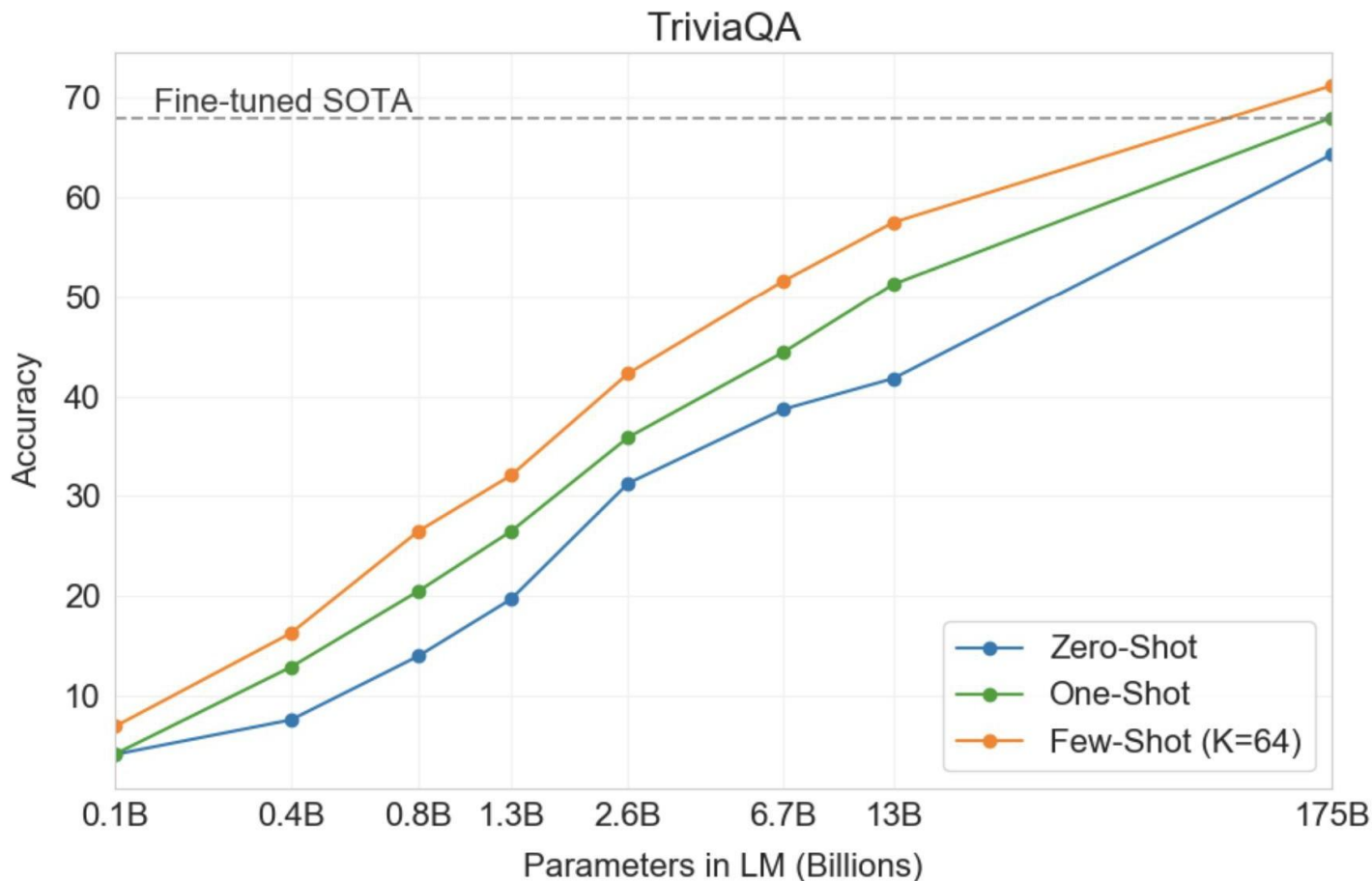
下游任务

- 上下文学习（零样本/小样本学习）



下游任务

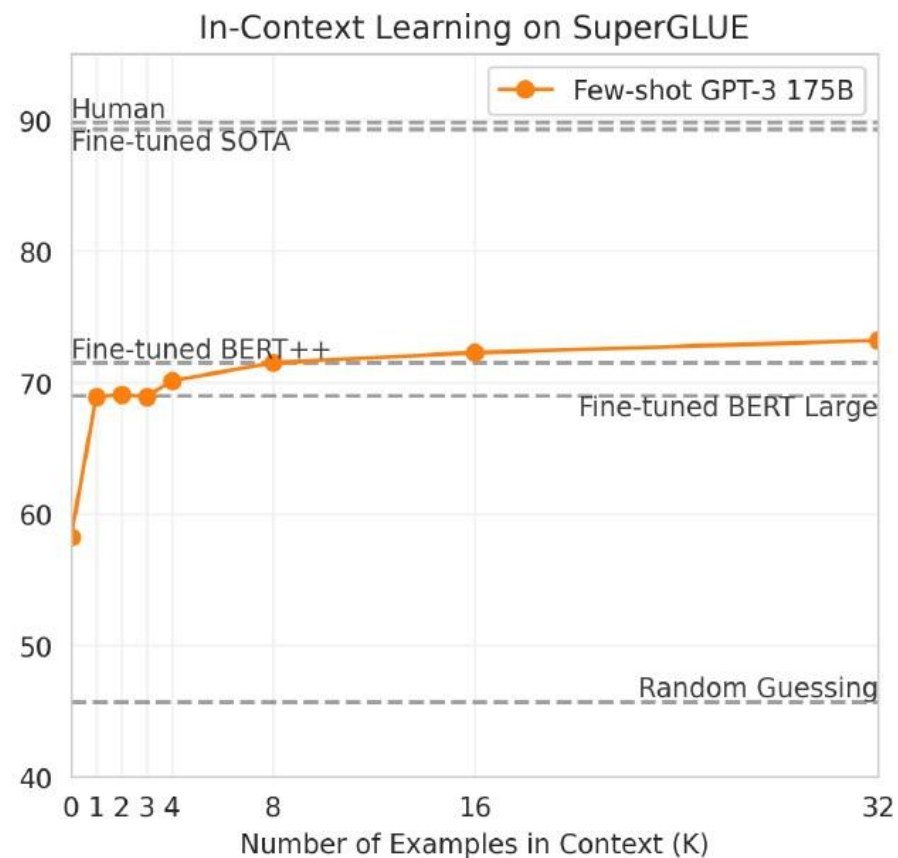
- 上下文学习（零样本/小样本学习）



Closed Book QA

下游任务

- 上下文学习（零样本/小样本学习）

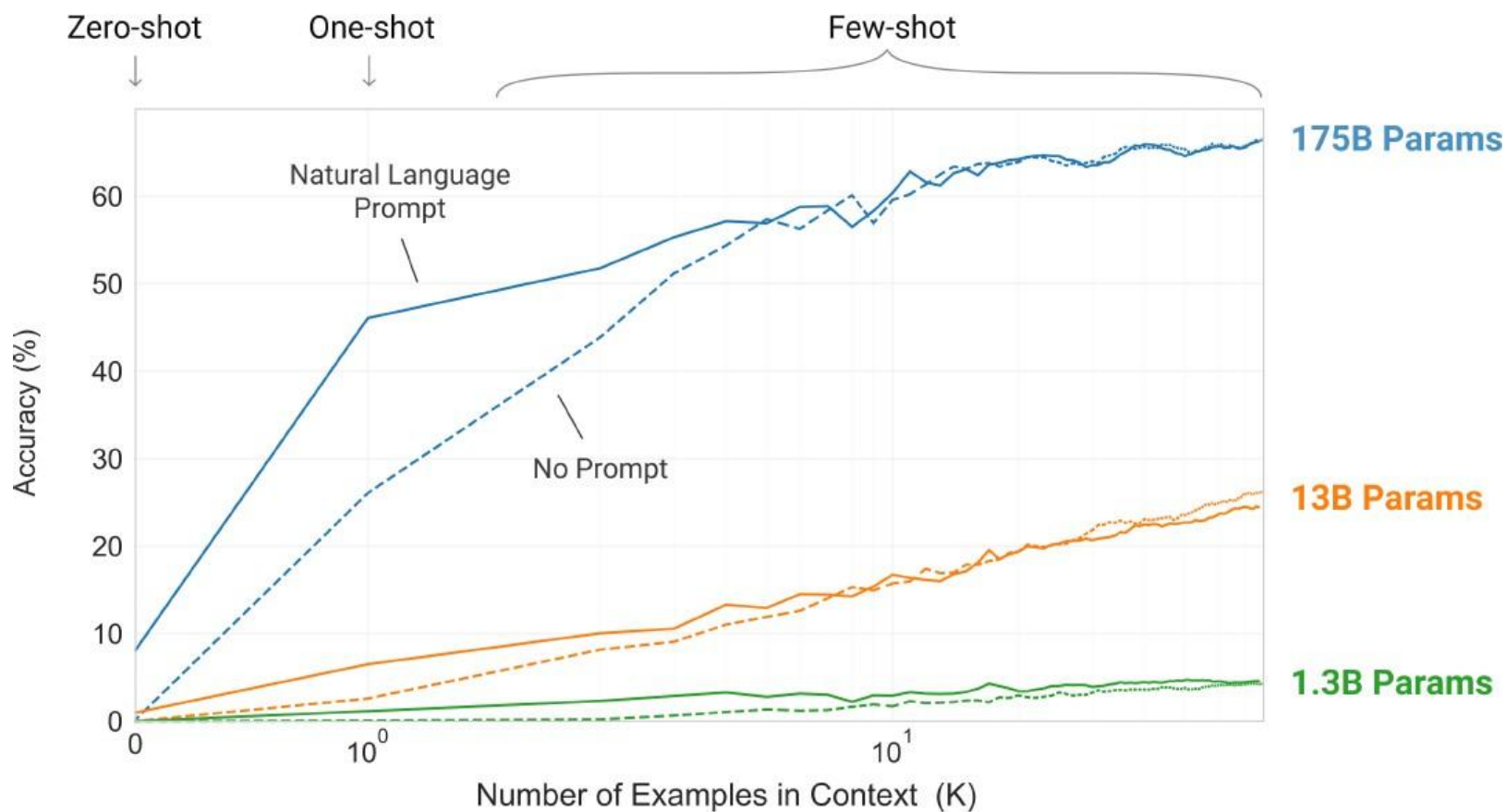


下游任务

- 上下文学习（零样本/小样本学习）

Context → Please unscramble the letters into a word, and write that word:
r e ! c . i p r o . c a / l =

Target Completion → reciprocal

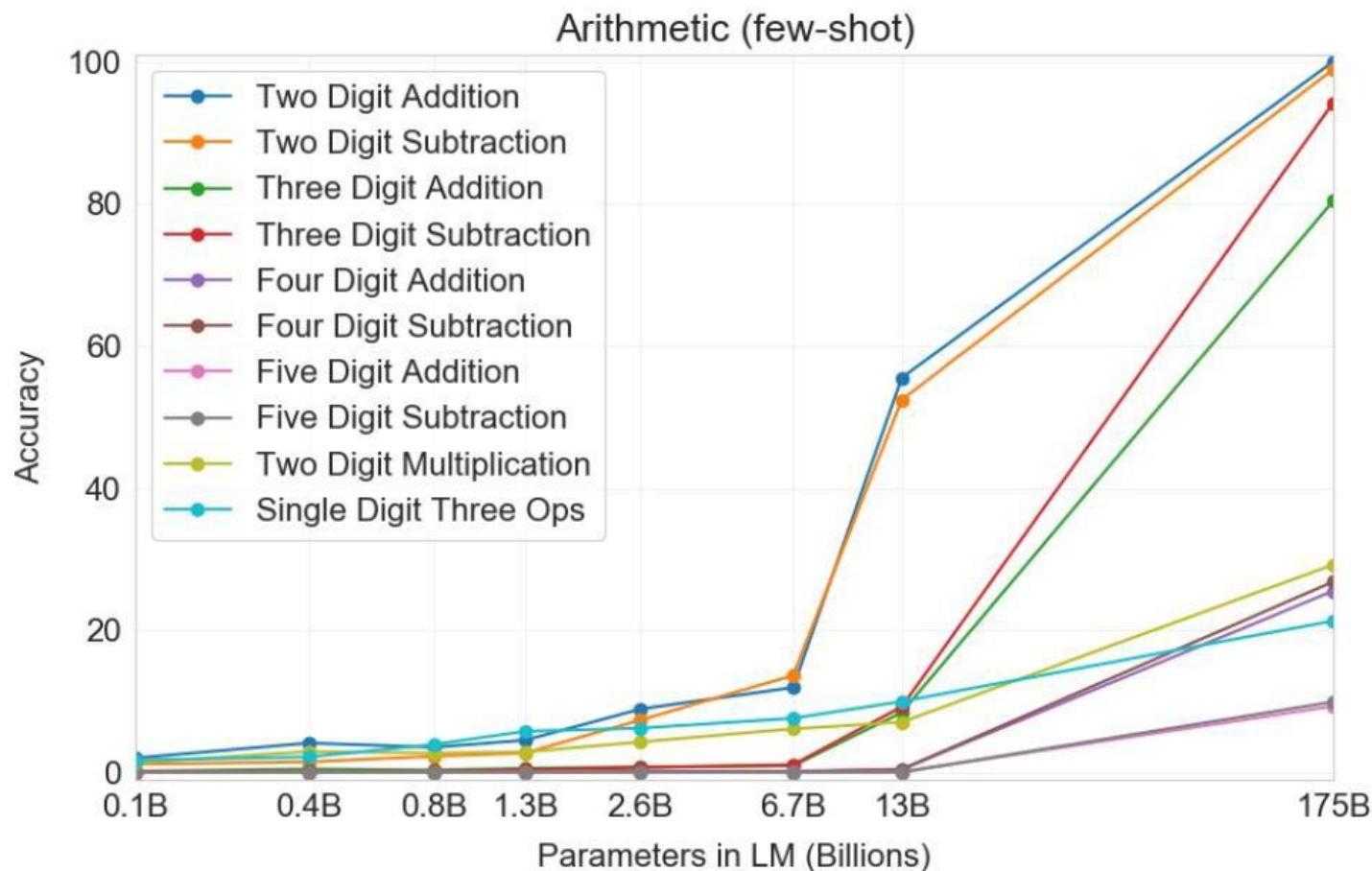


下游任务

- 上下文学习（零样本/小样本学习）

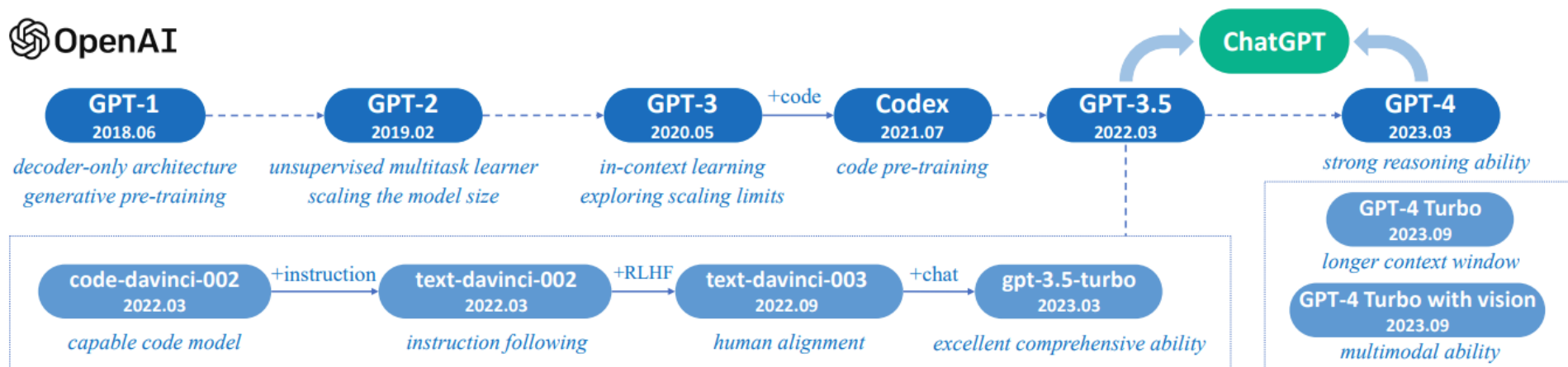
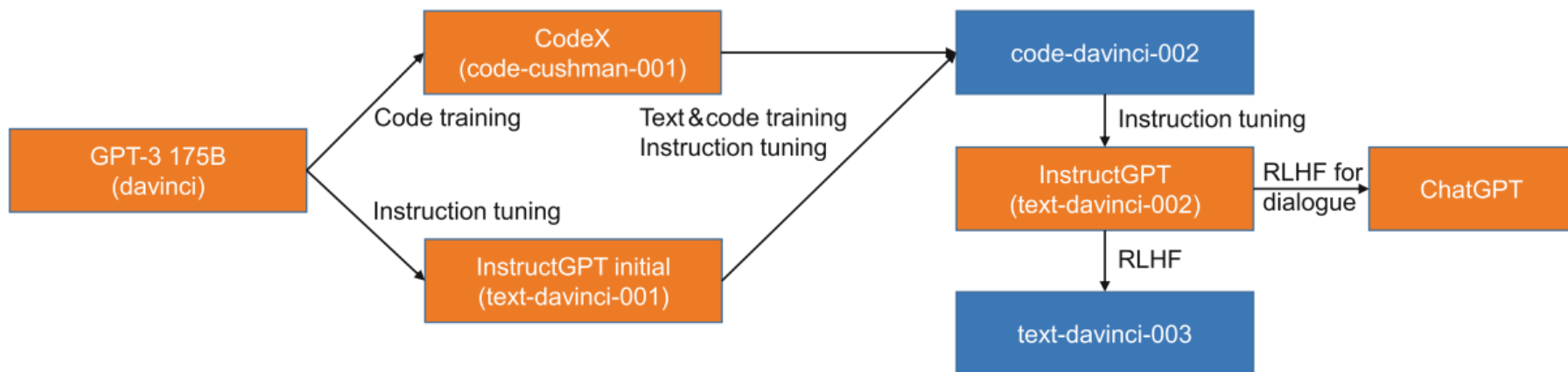
Context → Q: What is 17 minus 14?
A:

Target Completion → 3



ChatGPT模型迭代历程

- 从GPT-3到ChatGPT



ChatGPT模型效果

- ChatGPT

 详细介绍一下深度学习



深度学习（Deep Learning）是一种基于人工神经网络模型进行机器学习的方法，其特点是具有多层次的非线性特征提取和抽象能力。深度学习已经在各种应用领域取得了很好的效果，包括语音识别、图像识别、自然语言处理、游戏等领域。

深度学习的基本思想是使用多层次的神经网络对输入数据进行非线性变换和特征提取，然后通过训练过程优化网络参数，以实现分类、回归、聚类等任务。与传统的浅层神经网络相比，深度学习的多层次结构能够更好地捕捉数据的抽象特征和表示，从而提高模型的准确性和泛化能力。

深度学习主要包括以下几个方面的内容：

- 神经网络模型：神经网络是深度学习的核心模型，包括全连接神经网络、卷积神经网络、循环神经网络等多种类型，每种类型的神经网络都有不同的结构和应用场景。
- 模型训练：深度学习模型的训练通常使用反向传播算法进行，该算法能够自动计算网络参数的梯度，从而实现参数的优化。
- 模型评估与调优：深度学习模型的评估通常使用交叉验证等方法，调优则需要选择合适的网络结构、损失函数、优化算法等参数。
- 应用领域：深度学习已经被广泛应用于语音识别、图像识别、自然语言处理、游戏等领域，取得了很好的效果。

总的来说，深度学习是一种强大的机器学习方法，能够有效地解决许多复杂的数据分析和处理任务，对未来的发展具有广阔的应用前景。

ChatGPT模型效果

- ChatGPT

 你可以教我深度学习吗？



当然可以！深度学习是一种强大的机器学习方法，已经在很多领域取得了重大的突破，包括计算机视觉、自然语言处理、语音识别等。以下是一些深度学习的基础知识和资源，希望能对你有所帮助。

1. 什么是深度学习？

深度学习是一种基于人工神经网络的机器学习方法，其灵感来源于生物神经系统。深度学习的核心思想是通过大规模的数据训练神经网络，从而使其具备学习能力，从而能够解决很多实际问题。深度学习的特点是具有多个隐藏层，因此可以学习到更加复杂的特征。

2. 深度学习的基础知识

深度学习的基础知识包括神经网络、反向传播算法、激活函数、损失函数等。

神经网络

神经网络是深度学习的基础模型，它由多个神经元（或节点）组成，每个神经元接收一些输入，进行一些简单的运算，然后将输出传递给其他神经元。多个神经元可以组成一个层，多个层可以组成一个神经网络。

反向传播算法

反向传播算法是一种基于梯度下降的优化算法，它可以通过计算神经网络中每个参数对损失

ChatGPT模型效果

- ChatGPT



这些内容好多呀，我应该先学哪些内容呢？



学习深度学习需要一些预备知识，如果您还没有接触过以下内容，建议您先学习：

- Python 编程语言和基本的数据结构和算法
- 线性代数和概率论基础
- 机器学习基础，包括监督学习、无监督学习、强化学习等

如果您已经掌握了以上基础，那么您可以开始学习深度学习的基本理论和常用深度学习框架，如 TensorFlow、PyTorch 等，并通过实践来加深自己的理解和掌握深度学习技能。

在学习深度学习的过程中，建议您注重理论和实践相结合，通过不断地阅读论文和实践项目来加深理解和掌握技能。此外，也可以加入深度学习相关的社区和论坛，与其他爱好者交流

ChatGPT训练过程（方法）

- ChatGPT
 - OpenAI 未发表ChatGPT论文，训练过程与Instruct GPT类似

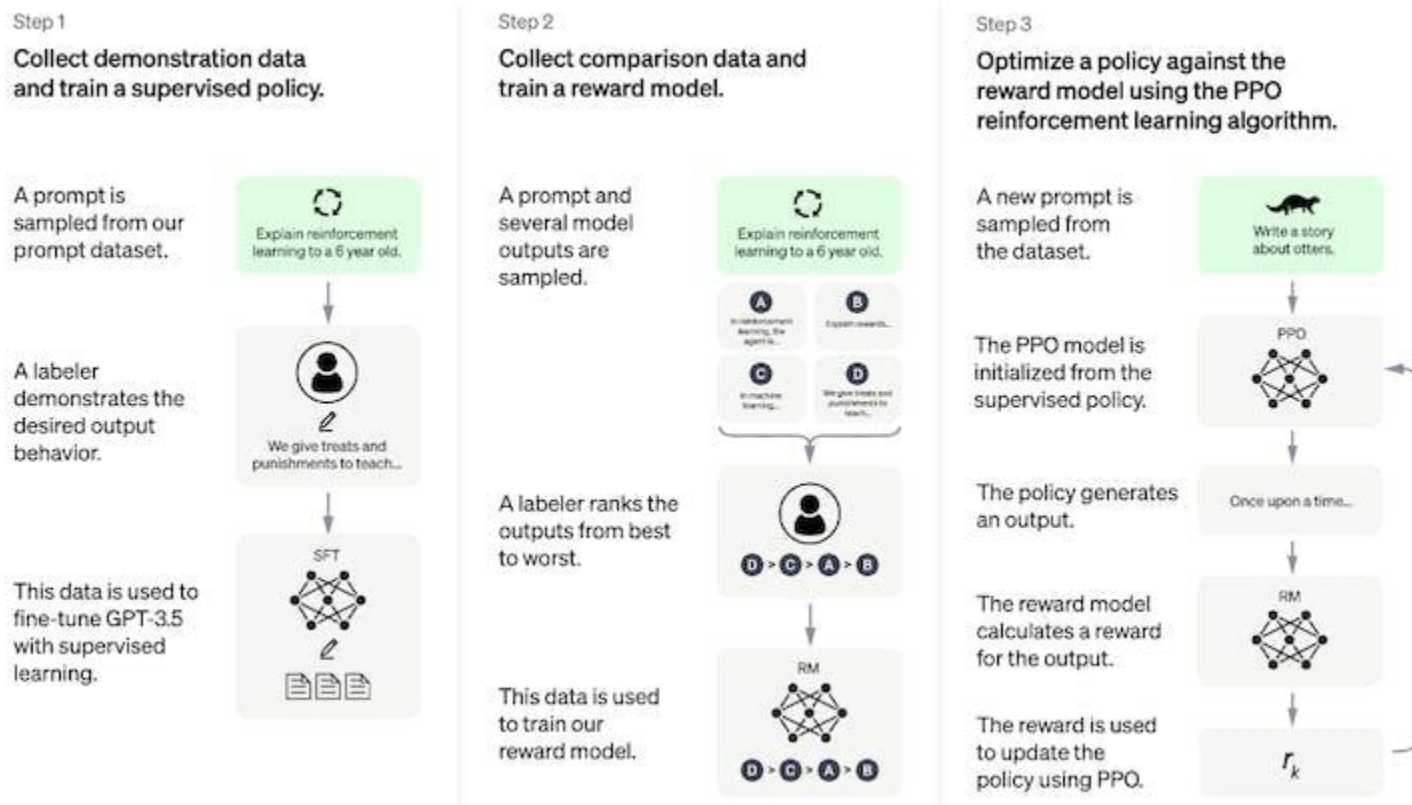


36th Conference on Neural Information Processing Systems (NeurIPS 2022).

ChatGPT训练过程（方法）

- ChatGPT

- OpenAI 未发表ChatGPT论文，训练过程与Instruct GPT类似
- 人类反馈强化学习方法（Reinforcement Learning from Human Feedback, RLHF）



GPT-4

- GPT-4

GPT-4 Technical Report

OpenAI*

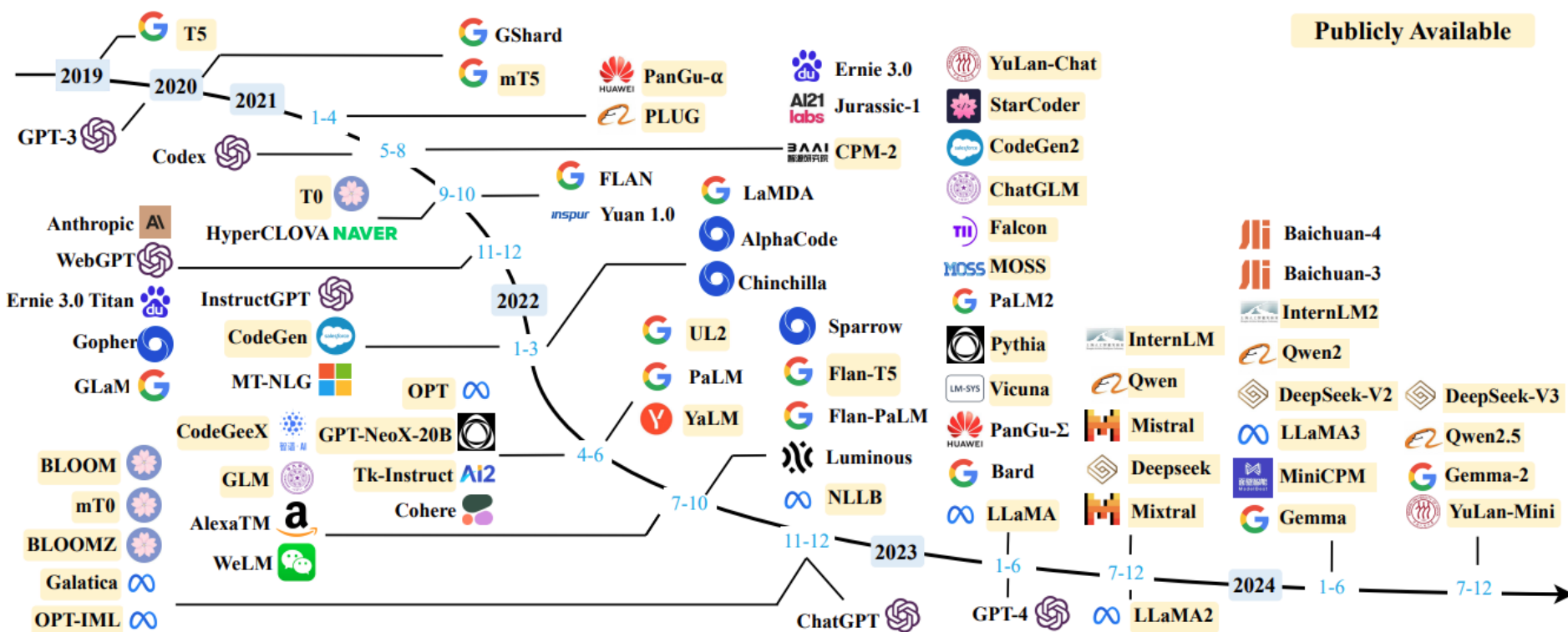
Abstract

We report the development of GPT-4, a large-scale, multimodal model which can accept image and text inputs and produce text outputs. While less capable than humans in many real-world scenarios, GPT-4 exhibits human-level performance on various professional and academic benchmarks, including passing a simulated bar exam with a score around the top 10% of test takers. GPT-4 is a Transformer-based model pre-trained to predict the next token in a document. The post-training alignment process results in improved performance on measures of factuality and adherence to desired behavior. A core component of this project was developing infrastructure and optimization methods that behave predictably across a wide range of scales. This allowed us to accurately predict some aspects of GPT-4's performance based on models trained with no more than 1/1,000th the compute of GPT-4.

16 Mar 2023

LLM整体发展进程

- 截至2024年为止的开源/闭源大语言模型



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 - 8.4.3 基于Transformer解码器的预训练语言模型
 - 8.4.4 基于Transformer的文本-视觉预训练模型
 - 预训练任务
 - 下游任务
- 8.5 本章小结

文本-视觉预训练数据



‘man with his dog on a couch’

文本-视觉预训练任务

- Masked Language Modeling (MLM)

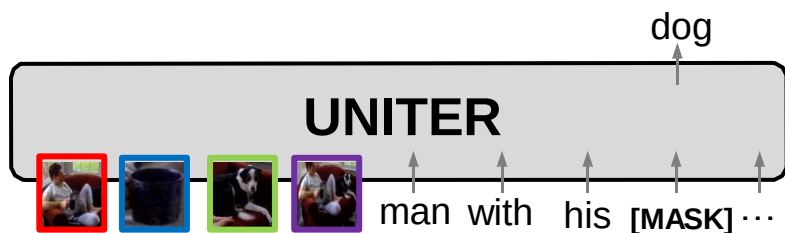


Image Regions: $\mathbf{v} = \{v_1, \dots, v_K\}$

Sentence Tokens: $\mathbf{w} = \{w_1, \dots, w_T\}$

Masking Indices: $\mathbf{m} \in \mathbb{N}^M$

Loss Function of Masked Language Modeling (MLM):

$$\mathcal{L}_{\text{MLM}}(\theta) = -E_{(\mathbf{w}, \mathbf{v}) \sim D} \log P_{\theta}(\mathbf{w}_{\mathbf{m}} | \mathbf{w}_{\setminus \mathbf{m}}, \mathbf{v}).$$

Reference:

1. Chen Y C, Li L, Yu L, et al. *Uniter: Universal image-text representation learning*. ECCV 2020

文本-视觉预训练任务

- Masked Region Modeling (MRM)

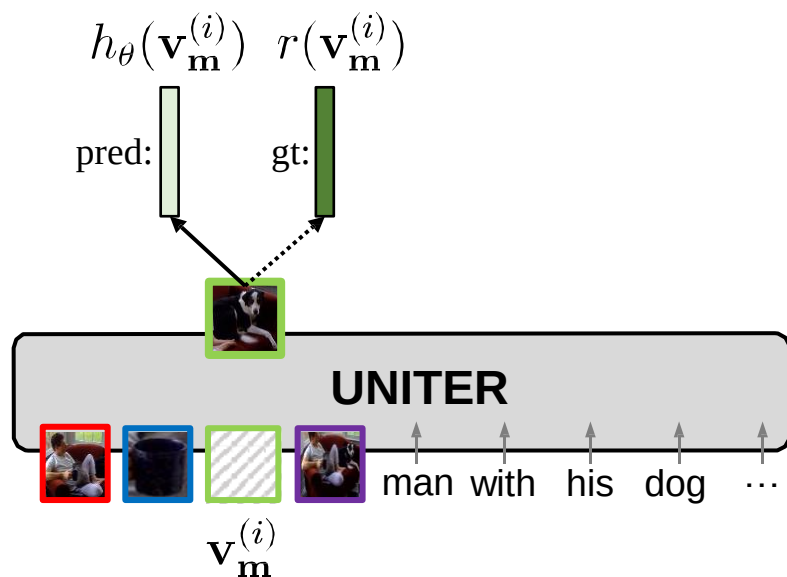


Image Regions: $\mathbf{v} = \{v_1, \dots, v_K\}$

Sentence Tokens: $\mathbf{w} = \{w_1, \dots, w_T\}$

Masking Indices: $\mathbf{m} \in \mathbb{N}^M$

Loss Function of Masked Region Modeling:

$$\mathcal{L}_{\text{MRM}}(\theta) = E_{(\mathbf{w}, \mathbf{v}) \sim D} f_\theta(\mathbf{v}_m | \mathbf{v}_{\setminus \mathbf{m}}, \mathbf{w}).$$

1) Objective of Masked Region Feature Regression (MRFR)

$$f_\theta(\mathbf{v}_m | \mathbf{v}_{\setminus \mathbf{m}}, \mathbf{w}) = \sum_{i=1}^M \|h_\theta(\mathbf{v}_m^{(i)}) - r(\mathbf{v}_m^{(i)})\|_2^2$$

Reference:

1. Chen Y C, Li L, Yu L, et al. *Uniter: Universal image-text representation learning*. ECCV 2020

文本-视觉预训练任务

- Masked Region Modeling (MRM)

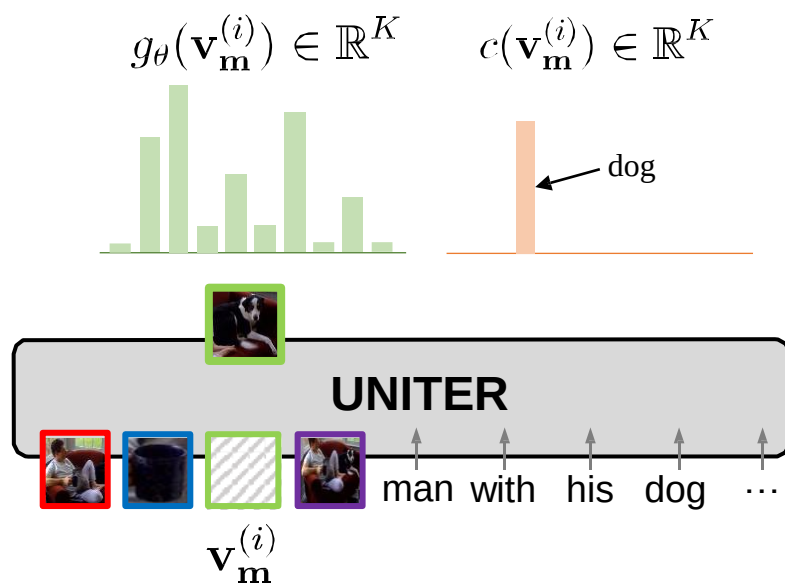


Image Regions: $\mathbf{v} = \{v_1, \dots, v_K\}$

Sentence Tokens: $\mathbf{w} = \{w_1, \dots, w_T\}$

Masking Indices: $\mathbf{m} \in \mathbb{N}^M$

Loss Function of Masked Region Modeling:

$$\mathcal{L}_{\text{MRM}}(\theta) = E_{(\mathbf{w}, \mathbf{v}) \sim D} f_{\theta}(\mathbf{v}_m | \mathbf{v}_{\setminus m}, \mathbf{w}).$$

2) Objective of Masked Region Classification (MRC)

$$f_{\theta}(\mathbf{v}_m | \mathbf{v}_{\setminus m}, \mathbf{w}) = \sum_{i=1}^M \text{CE}(c(\mathbf{v}_m^{(i)}), g_{\theta}(\mathbf{v}_m^{(i)}))$$

Reference:

1. Chen Y C, Li L, Yu L, et al. *Uniter: Universal image-text representation learning*. ECCV 2020

文本-视觉预训练任务

- Masked Region Modeling (MRM)

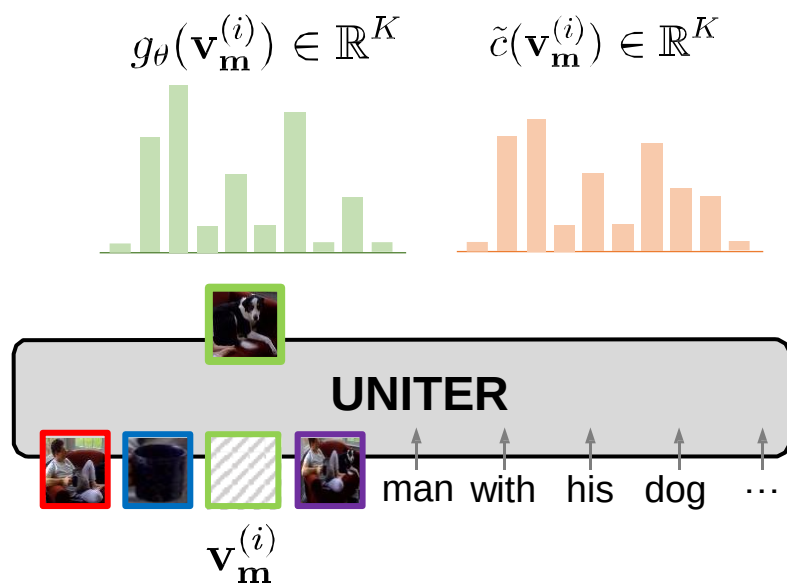


Image Regions: $\mathbf{v} = \{v_1, \dots, v_K\}$

Sentence Tokens: $\mathbf{w} = \{w_1, \dots, w_T\}$

Masking Indices: $\mathbf{m} \in \mathbb{N}^M$

Loss Function of Masked Region Modeling:

$$\mathcal{L}_{\text{MRM}}(\theta) = E_{(\mathbf{w}, \mathbf{v}) \sim D} f_\theta(\mathbf{v}_m | \mathbf{v}_{\setminus \mathbf{m}}, \mathbf{w}).$$

3) Objective of **Masked Region Classification – KL Divergence (MRC-kl)**

$$f_\theta(\mathbf{v}_m | \mathbf{v}_{\setminus \mathbf{m}}, \mathbf{w}) = \sum_{i=1}^M D_{KL}(\tilde{c}(\mathbf{v}_m^{(i)}) || g_\theta(\mathbf{v}_m^{(i)}))$$

Reference:

1. Chen Y C, Li L, Yu L, et al. *Uniter: Universal image-text representation learning*. ECCV 2020

文本-视觉预训练任务

- Image-Text Matching (ITM)

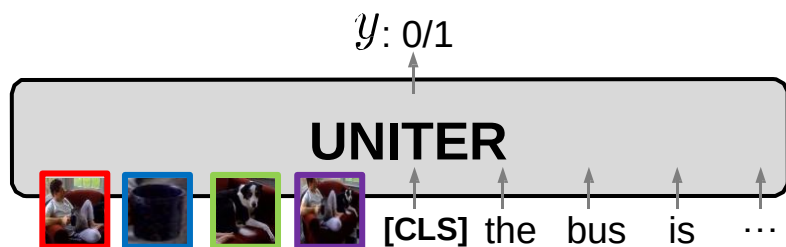


Image Regions: $\mathbf{v} = \{v_1, \dots, v_K\}$

Sentence Tokens: $\mathbf{w} = \{w_1, \dots, w_T\}$

Masking Indices: $\mathbf{m} \in \mathbb{N}^M$

Loss Function of **Image-Text Matching (ITM)**

$$\mathcal{L}_{\text{ITM}}(\theta) = -E_{(\mathbf{w}, \mathbf{v}) \sim D} [y \log s_{\theta}(\mathbf{w}, \mathbf{v}) + (1 - y) \log(1 - s_{\theta}(\mathbf{w}, \mathbf{v}))].$$

Reference:

1. Chen Y C, Li L, Yu L, et al. *Uniter: Universal image-text representation learning*. ECCV 2020

文本-视觉预训练任务

• CLIP模型-图文对比学习

Learning Transferable Visual Models From Natural Language Supervision

Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹

Abstract

SOTA computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study performance on over 30 different computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at <https://github.com/OpenAI/CLIP>.

1. Introduction and Motivating Work

Pre-training methods which learn directly from raw text have revolutionized NLP over the last few years (Dai & Le, 2015; Peters et al., 2018; Howard & Ruder, 2018; Radford et al., 2018; Devlin et al., 2018; Raffel et al., 2019). The development of “text-to-text” as a standardized input-output

interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Joulin et al. (2016) demonstrated that CNNs trained to predict words in image captions can learn representations competitive with ImageNet training. Li et al. (2017) then extended this approach to predicting phrase n-grams in addition to individual words and demonstrated the ability of their system to zero-shot transfer to other image classification datasets. Adopting more recent architectures and pre-training approaches, ViT (Dosovitskiy & Johnson, 2020), ICMLM (Bulent Sariyildiz et al., 2020), and ConVIRT (Zhang et al., 2020) have recently demonstrated the potential of transformer-based language modeling, masked language modeling, and contrastive objectives to learn image representations from text.

However, the aforementioned models still under-perform current SOTA computer vision models such as Big Transfer (Kolesnikov et al., 2019) and the weakly supervised ResNeXt (Mahajan et al., 2018). A crucial difference is scale. While Mahajan et al. (2018) and Kolesnikov et al. (2019) trained for accelerator years on millions to billions of images, ViT, ICMLM, and ConVIRT trained for accelerator days on one to two hundred thousand images. We close this gap and study the behaviors of image models trained from natural language supervision at large scale. We demonstrate that a simplified version of ConVIRT trained from scratch, which we call CLIP, for Contrastive Language-Image Pre-training, is an efficient and scalable method of learning from natural language supervision. We find that

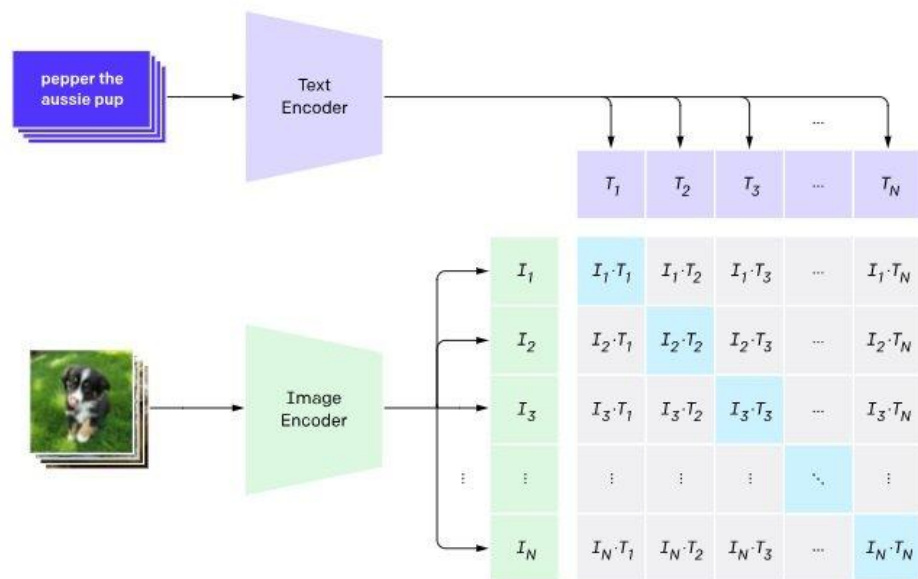
^{*}Equal contribution ¹OpenAI, San Francisco, CA 94110, USA. Correspondence to: <{alec, jongwook}@openai.com>.

Proceedings of the 38th International Conference on Machine Learning, PMLR 139, 2021. Copyright 2021 by the author(s).

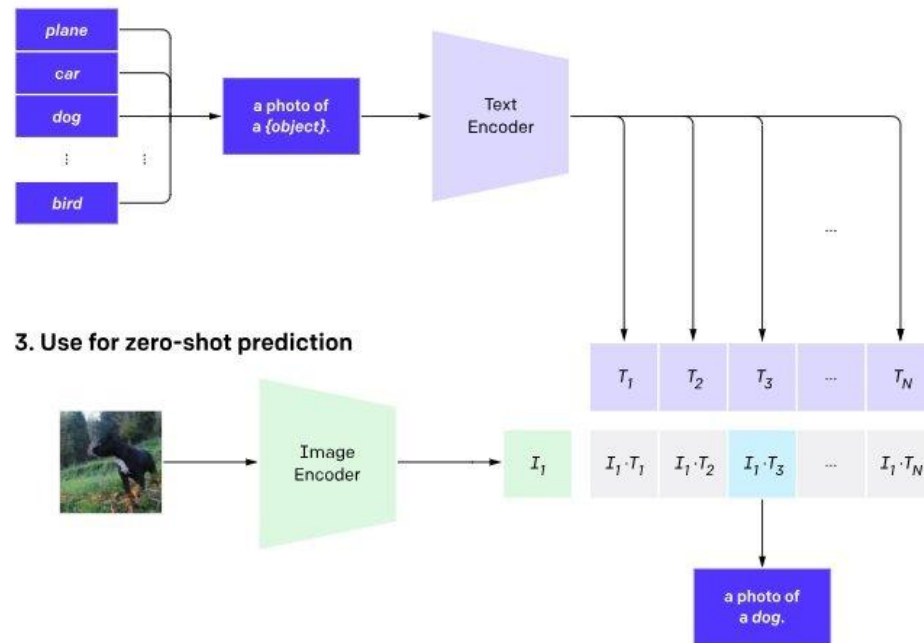
文本-视觉预训练任务

• CLIP模型-图文对比学习

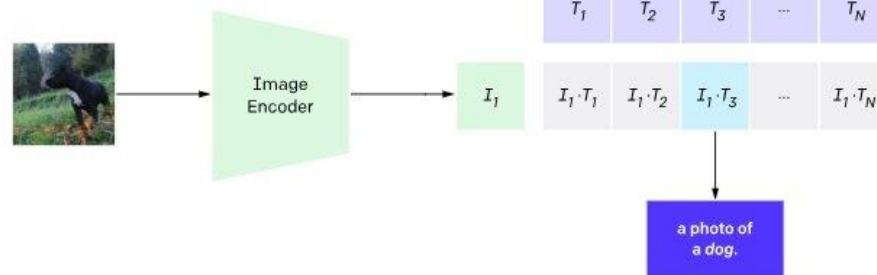
1. Contrastive pre-training



2. Create dataset classifier from label text



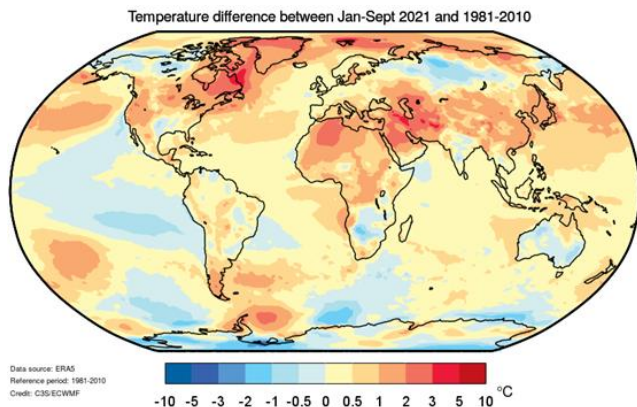
3. Use for zero-shot prediction



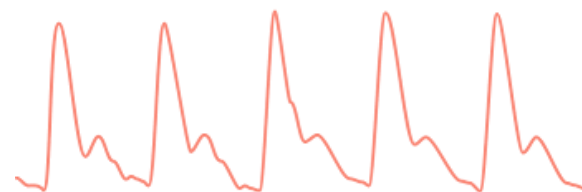
多模态大模型

- 动机

- 真实物理世界中有文本、视觉、音频、3D、雷达、多谱等复杂多样的不同模态信号



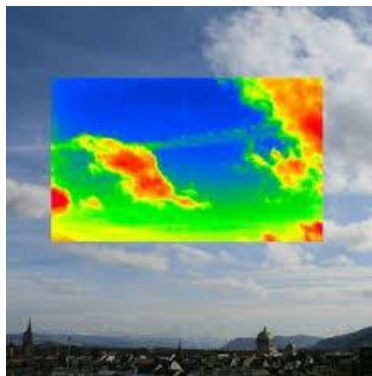
温度



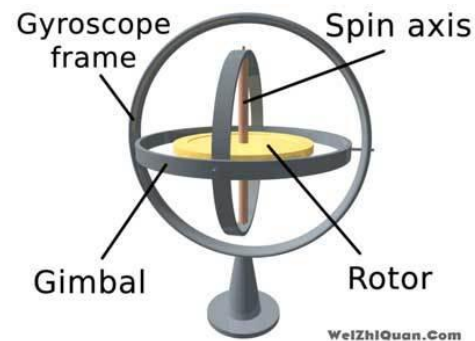
脉搏波



3D超声波指纹



红外信号



陀螺仪

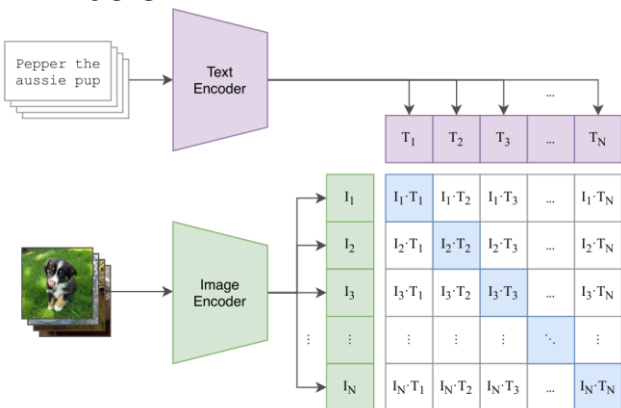
多模态大模型

三类范式

面向特定任务的多模态大模型

面向跨模态检索和生成等特定任务构建的多模态大模型

例如CLIP、DALL-E、Stable Diffusion



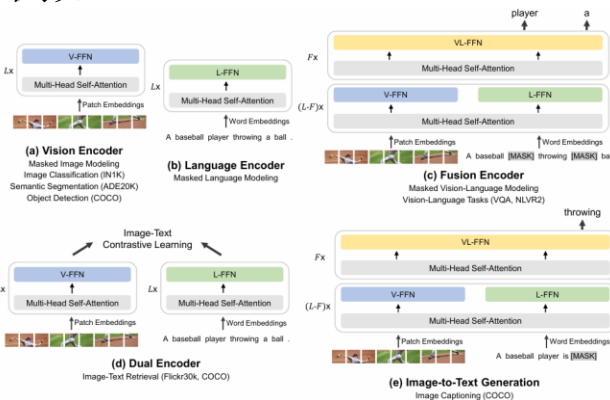
特点:

1. 训练任务单一
2. 模型参数规模较小(千万、亿级参数)
3. 下游任务受限

多任务多模态大模型:

基于单一模型架构, 通过若干跨模态任务同时训练产生的多模态大模型

例如SAM、BEiT-3



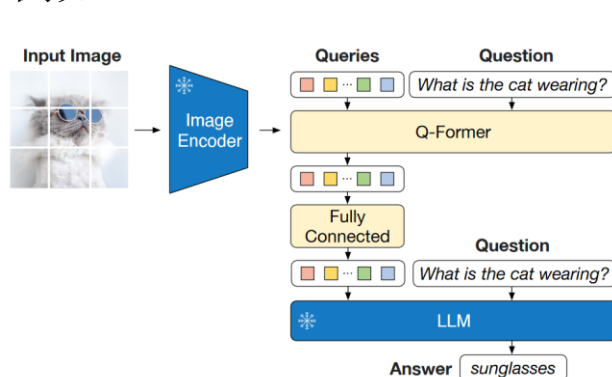
特点:

1. 训练任务受限, 多为监督数据
2. 模型参数规模较小(十亿级参数)
3. 通用能力弱

多模态大语言模型:

以大语言模型未认知基座, 将其他各种模态对齐到语言空间, 实现多模态大模型

例如BLIP-2、LLaVA



特点:

1. 综合自监督、有监督数据, 数据量大
2. 模型参数规模大(百亿、千亿级参数)
3. 通用能力强

多模态大模型发展历程

• 截止2024年3月

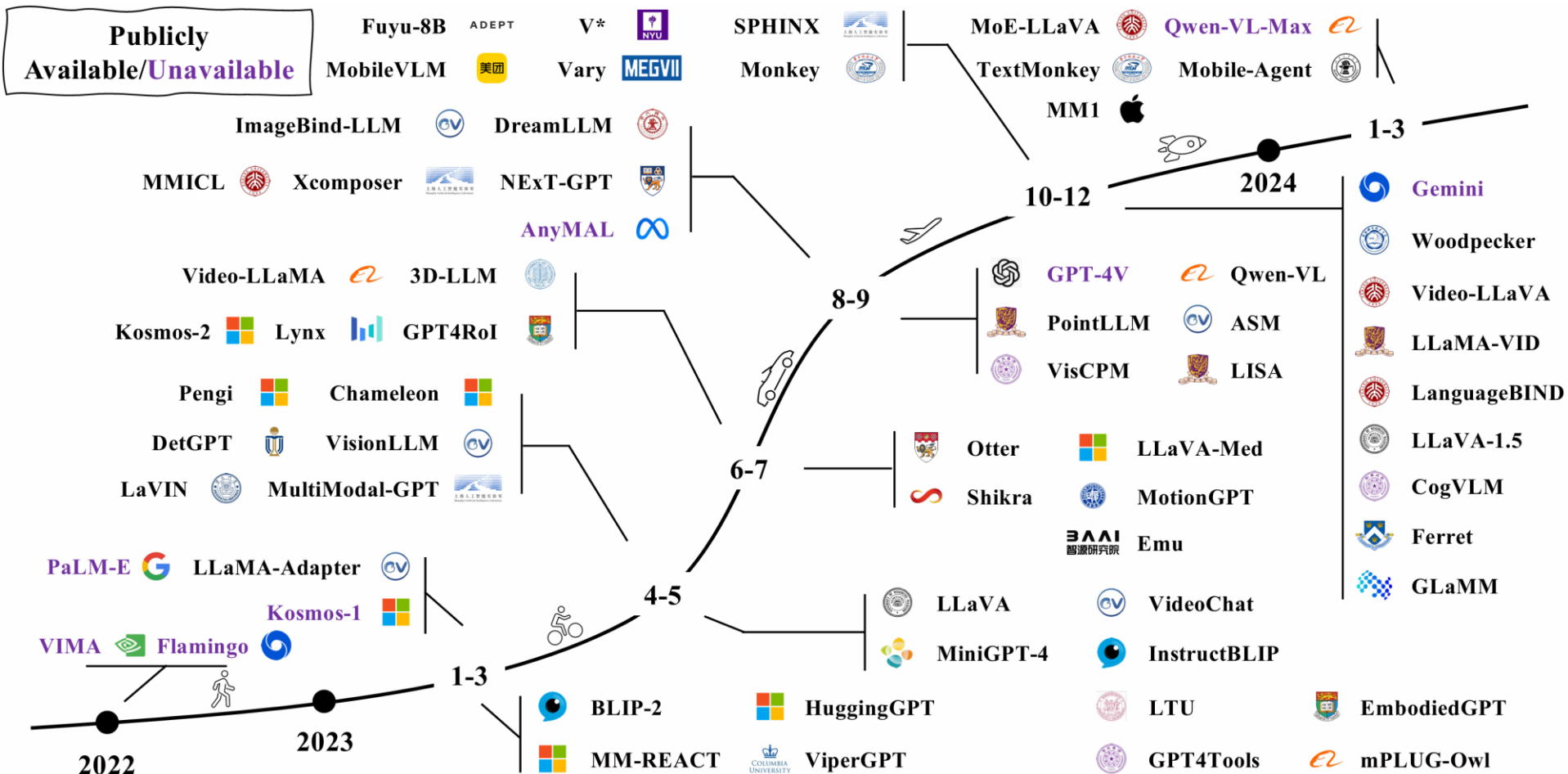


Fig. 1: A timeline of representative MLLMs. We are witnessing rapid growth in this field. More works can be found in our released GitHub page, which is updated daily.

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- 8.4 预训练模型及其应用
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 - 8.4.2 基于Transformer的预训练Seq2Seq模型
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 - 8.4.4 基于Transformer的文本-视觉预训练模型
 - 预训练任务
 - 下游任务
- 8.5 本章小结

文本-视觉下游任务

- Visual Question Answering

Result



Select Task



Input Field

Submit

<https://vilbert.cloudcv.org/>

文本-视觉下游任务

- **Spatial Visual Reasoning**

Result



Select Task



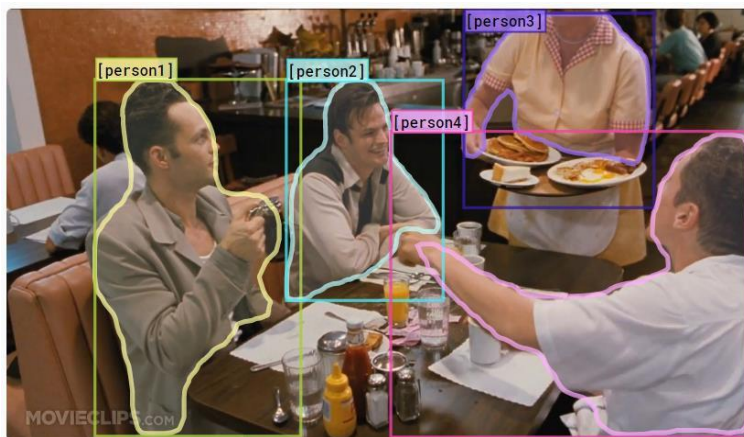
Input Field

Submit

<https://vilbert.cloudcv.org/>

文本-视觉下游任务

• Visual Commonsense Reasoning



Why is [person4] pointing at [person1]?

- a) He is telling [person3] that [person1] ordered the pancakes.
- b) He just told a joke.
- c) He is feeling accusatory towards [person1].
- d) He is giving [person1] directions.

I choose (a) because:

- a) [person1] has the pancakes in front of him.
- b) [person4] is taking everyone's order and asked for clarification.
- c) [person3] is looking at the pancakes and both she and [person2] are smiling slightly.
- d) [person3] is delivering food to the table, and she might not know whose order is whose.

Reference:

1. *From Recognition to Cognition: Visual Commonsense Reasoning*, CVPR 2019

文本-视觉下游任务

- Referring Expression Comprehension



woman washing dishes

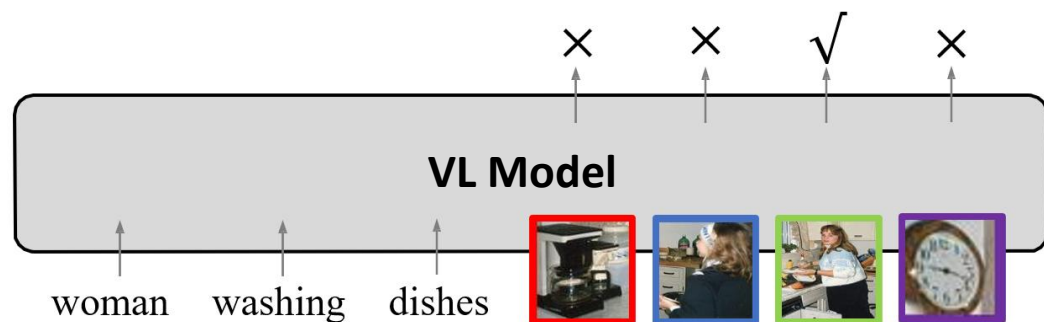
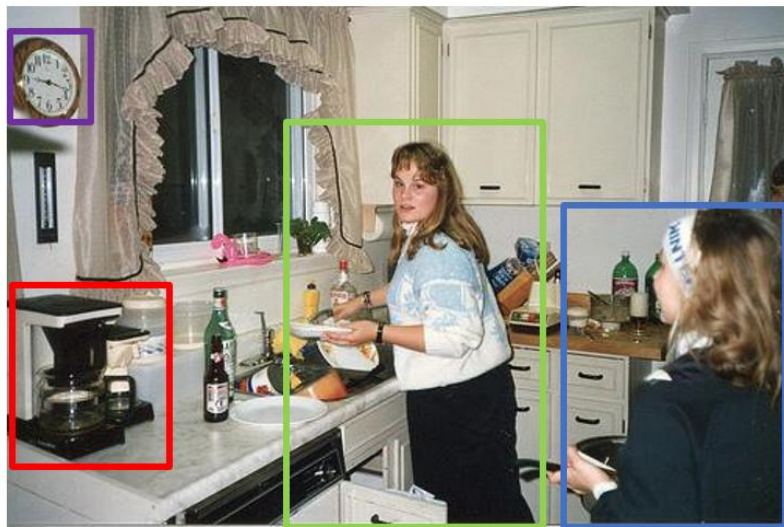


Reference:

1. Flickr30K Entities: Collecting Region-to-Phrase Correspondences for Richer Image-to-Sentence Models, *IJCV 2017*

文本-视觉下游任务

- Referring Expression Comprehension



文本-视觉下游任务

- Referring Expression Comprehension

Result



Select Task

▼

Input Field

Submit

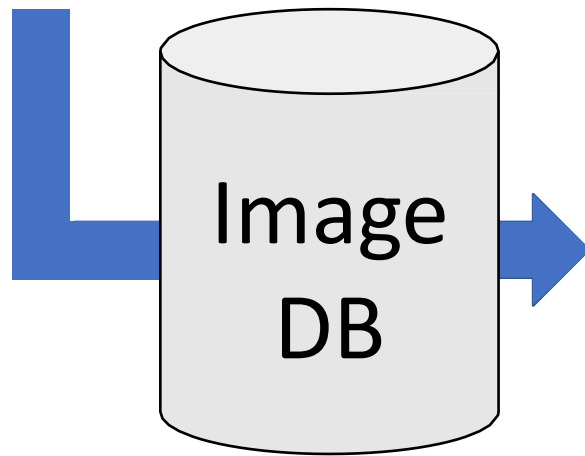
Credits

<https://vilbert.cloudcv.org/>

文本-视觉下游任务

- Image-Text Retrieval

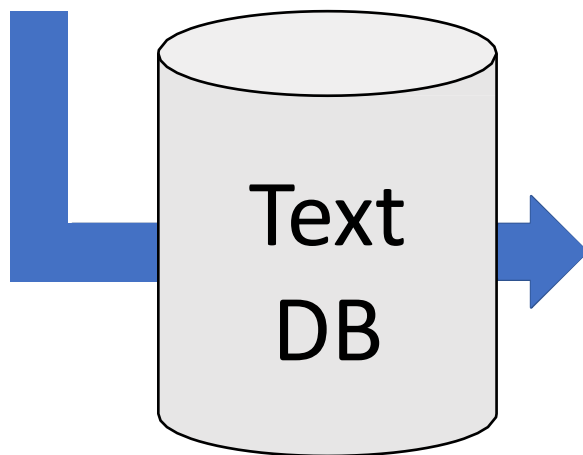
“a girl with a cat on grass”



...

文本-视觉下游任务

- Image-Text Retrieval



- “four people with ski poles in their hands in the snow”
- “four skiers hold on to their poles in a snowy forest”
- “a group of young men riding skis”
- “skiers pose for a picture while outside in the woods”
- “a group of people cross country skiing in the woods”

• • •

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- 8.5 本章小结



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